

Member-specific thresholds and contested-price intervals for stage-gate screening: A curvature-based real-options method under regulatory ambiguity

Abstract

Stage-gate committees decide when to make irreversible investments under regulatory ambiguity, and members weighting adverse scenarios differently can reach opposite decisions at the same price. Real-options analysis reduces this decision to one project-level trigger, and committee decision support compresses member assessments into a score or ranking, so neither reports the price range over which member classifications differ. This study therefore derives one price threshold per member from affine combinations of the triggers at both ends of a volatility interval, so a committee can locate disagreement before any vote. The extreme thresholds bound a contested-price interval, and a unique critical weight splits members around the no-ambiguity anchor. Under the affine rule, trigger curvature places this weight above one half in standard-deviation units and below it in variance units. Against a pre-commitment α -maxmin expected utility benchmark, the largest relative trigger error over the primary radii was 2.74%. Because small trigger errors did not guarantee identical classifications, affine-only release is capped at 10%, the largest evaluated radius at which two fidelity criteria held. In Monte Carlo committee panels, the member-threshold vector retained investment-timing information, with 88.5% lower median loss than the ex post best of four single-trigger summaries. In a carbon capture, utilization, and storage retrofit, all archived prices stayed below the lower edge under the reference configuration. In 15 of 32 sensitivity configurations, however, the highest archived price reached or exceeded this edge. Unanimity at a gate therefore depended on the recorded inputs, and the member-level report made this dependence visible.

Keywords. Real options, Stage-gate screening, Regulatory ambiguity, Group decision support, CCUS investment

1. Introduction

Large, irreversible investments are commonly reviewed through a sequence of stage gates (Cooper, 2008). At each gate, a committee decides whether to commit to the project now or to defer it to a later review. Real-options analysis reduces this timing decision to a single investment trigger (McDonald and Siegel, 1986; Dixit and Pindyck, 1994). Commitment is warranted once project value reaches this trigger. A single trigger gives the review one compact criterion, which can be documented and recomputed at successive gates. When project cash flows are linked to an observable price, the trigger can be restated on the price axis, in units the committee already monitors. In this paper, a trigger is a project-value quantity, and a threshold is its image on the price axis.

Committee members, however, may receive the same engineering, cost, and policy evidence and still place different weights on adverse regulatory scenarios. An ambiguity-aware real-options analysis returns one project-level trigger under a representative ambiguity attitude, so member-level dispersion disappears at the modeling stage. Committee decision-support procedures can retain individual assessments, but they aggregate them into a score, a ranking, or a vote before reporting. In both cases, the assessments are compressed before their implied classifications are compared.

This compression matters for the gate only when member-level dispersion changes the classification, which does not always happen. All members receive a wait classification at a price below every member threshold and an invest classification at a

price at or above every threshold. A mixed classification arises only between the lowest and the highest member thresholds, which bound the contested-price interval. The position of the price relative to the whole threshold set therefore determines whether one classification suffices. An aggregate score replaces this set with one number and thus hides whether the price lies between the extreme thresholds. The set, however, remains recoverable from the recorded calibration, ambiguity radius, and member weights, and changing these inputs can reclassify the same prevailing price.

This paper determines when member heterogeneity changes the current gate classification and reports the answer without aggregating member thresholds in advance. All members share one ordered ambiguity set and differ only in their recorded weights on its adverse endpoint. Member triggers follow a closed-form affine endpoint rule, and the α -maxmin expected utility (α -MEU) optimum under the same inputs serves as its accuracy benchmark. The two rules can classify the same price differently. The paper therefore also determines when the affine report can be released alone and when the benchmark must be recomputed.

The shared ambiguity set represents one component of regulatory uncertainty as an ordered interval of forward volatilities, with a declared radius around an anchor. Its adverse endpoint is the low-volatility endpoint, because lower volatility reduces the value of the option to wait and lowers the trigger.

The affine rule also determines whether each member trigger lies above or below the no-ambiguity anchor. The trigger is strictly increasing in volatility, so at a positive radius the anchor

trigger lies strictly between the two endpoint triggers. Exactly one weight in $[0, 1]$, called the critical weight, therefore places a member trigger at the anchor. This weight splits the members into two groups around the anchor, and the split carries over to the price thresholds. The critical weight is located relative to one half, the weight at which the two endpoint triggers count equally.

The running example is a committee screening a carbon capture, utilization, and storage (CCUS) retrofit of a coal-fired power plant. The retrofit costs 1.585×10^9 CNY and captures 1.69×10^6 tonnes of CO_2 per year (Sun et al., 2024). Project cash flows track an observable carbon price, so the member triggers become marginal carbon-price thresholds in Chinese Yuan per tonne (CNY/t). On the input side, the ambiguity radius of each gate comes from a policy-text series.

The paper makes one main contribution and two supporting contributions. The main contribution is a member-indexed gate report, which consists of the member-threshold vector and the contested-price interval bounded by its extremes. The report provides a pre-vote test of whether the prevailing price receives the same classification under every member assessment. The two supporting contributions are analytical and methodological. The analytical contribution is a location rule for the critical weight. Strict monotonicity of the affine rule makes this weight unique, and trigger curvature places it relative to one half. This curvature changes sign between standard-deviation and variance coordinates, so the direction of shift for members weighted near one half depends on the declared coordinate. The methodological contribution is a release rule for the affine report. Benchmark comparisons and a classification-fidelity analysis measure the accuracy of this report in price units. In this analysis, small trigger errors could still change the invest-or-wait outcome. A finite registry safeguard converts these accuracy limits into conditions for affine-only release, and each release can then be audited against the registered configurations.

The CCUS application produced two main results. First, unanimity depended on the configuration. Under the reference configuration, every archived price stayed below the contemporaneous lower interval edge. By contrast, in 15 of the 32 sensitivity configurations, the highest archived price reached or exceeded this edge. Second, the common appraisal inputs moved the interval edges further than member disagreement did. A 10% reduction in the cost-to-volume ratio I/Q moved the lower edge by 11.34 CNY/t. By comparison, the widest interval under the reference profile along the reconstructed path was 6.63 CNY/t.

By treating member disagreement as a reported quantity with stated accuracy limits, the study addresses a reporting gap between real-options analysis and committee decision support. This gap separates a trigger computed for the project from a decision taken by its committee.

2. Literature review

A gate report built from member thresholds and their contested-price interval relates to three research streams. Ambiguity-aware real options provide the trigger analysis, committee decision support provides the reporting context, and

text-based uncertainty measures provide one application input. Each stream supplies a different object to the gate and leaves one gap for member-level screening.

2.1. Real-options analytics for irreversible investment

McDonald and Siegel (1986) demonstrated the positive value of waiting under irreversibility and uncertainty. Dixit and Pindyck (1994) developed this result into the trigger analysis used here. Nadarajah and Secomandi (2023) surveyed the uptake of this framework in the operations literature on energy assets. In this literature, the valuation target is an operational asset, and the flexibility is managerial. The retrofit examined here is such an asset, and the quantity of interest is the date of commitment. In the standard perpetual geometric Brownian motion (GBM) model, this date depends on volatility, because higher volatility raises the value of waiting and the trigger. When volatility is ambiguous in the sense of Ellsberg (1961), more than one law for the price process remains admissible. The trigger then depends on how this ambiguity is represented and evaluated.

Nishimura and Ozaki (2007) placed Knightian ambiguity on the drift of the price process in an irreversible investment problem. They evaluated it under the maxmin expected utility (MEU) preferences of Gilboa and Schmeidler (1989). In their model, the worst-case prior reduces the value of waiting and accelerates commitment. Trojanowska and Kort (2010) extended the drift-ambiguity analysis to a finite project life, where acceleration held only in the short run. The α -maxmin generalization (Ghirardato et al., 2004) assigns weight α to the worst-case prior and $1 - \alpha$ to the best-case prior, nesting MEU at $\alpha = 1$. Under this broader formulation, Schröder (2011) showed how the investment decision depends on both the ambiguity and the attitude toward it. Depending on this attitude, the subjective project value can rise, and commitment can become more attractive. Two decision makers facing the same objective situation may therefore act differently. Schröder (2020) noted a further issue: ambiguity models of this class, including α -MEU, are generally not dynamically consistent, a property needed for recursive optimal stopping. The direction of the effect of ambiguity on exercise timing also varies with the setting. In Miao and Wang (2011), ambiguity could accelerate or delay option exercise, depending on whether it affected continuation payoffs, termination payoffs, or both. Drift and volatility enter the characteristic equation through different terms, so the comparative statics for drift ambiguity require rederivation under volatility ambiguity. Given the dynamic inconsistency of α -MEU, the benchmark in this paper is a pre-commitment rule. It is compared with a recursive alternative, which is time-consistent by construction.

Other formulations place ambiguity on volatility or on the model more broadly, and the decision question of each determines the object it returns. Four objects recur, and one formulation may return more than one of them. Formulations concerned with the worth of a claim return a value. The uncertainty-volatility model of Avellaneda et al. (1995) bounds the price of a claim by evaluating it at the extremes of a volatility band. The G -expectation calculus of Peng (2019) gives this band a coherent dynamic form, and Epstein and Ji (2013) carried the

Nomenclature

Decision and valuation variables

I	Investment (retrofit) cost
V^*	Baseline investment trigger, a project-value quantity, Eq. (2)
$V_{\text{low}}^*, V_{\text{high}}^*$	Endpoint triggers at the lower and upper volatility endpoints of the ambiguity set Σ_t
$V_{\text{lin}}^*, V_{\alpha\text{-MEU}}^*$	Member trigger under the affine rule and under the pre-commitment benchmark, at the same recorded weight
V_0^*	Anchor trigger, the trigger at the volatility anchor $\bar{\sigma}_0$
V_0	Reference project value at which the pre-commitment benchmark is evaluated
$V^*(\alpha, t)$	Member trigger under the affine rule, Eq. (5), written $V_{\text{lin}}^*(\alpha)$ in benchmark comparisons
P^*	Marginal carbon-price threshold, Eq. (8)
P_i^*	Member threshold, the marginal carbon-price threshold of member i , equal to P^* evaluated at α_i
$P_{i,\text{lin}}^*$	Marginal carbon-price threshold of member i under the affine rule, used by the safeguard distances
P_0^*	No-ambiguity marginal carbon-price threshold, called the anchor threshold
P_t	Prevailing effective carbon price at gate t
C_t	Contested-price interval, $[\min_i P_i^*, \max_i P_i^*]$. When all member thresholds coincide, this half-open interval is empty and $B_t = 0$
B_t	Width of the contested-price interval, Eq. (9)
τ	First hitting time of a candidate trigger in the benchmark problem
V_τ	Project value at the first hitting time τ
J^α	Pre-commitment α -MEU benchmark objective, Eq. (13)
c, s, Δ	Common per-tonne cost, common per-tonne subsidy, and their net adjustment $\Delta = c - s$
$L_{\text{lin}}, U_{\text{lin}}$	Lower and upper edges of the affine contested-price interval C_t , written L and U in the common-adjustment identity

Stochastic process and economic parameters

$\bar{\sigma}_0$	Forward-volatility anchor, $\bar{\sigma}_0 > 0$, estimated from historical carbon prices in the application
σ_p	Forward carbon-price volatility, the volatility parameter of the GBM price process
$\sigma_{\text{low}}, \sigma_{\text{high}}$	Lower and upper volatility endpoints of Σ_t
μ_c	Annual carbon-price growth scenario, declared as a decision input
r	Discount rate
δ	Net discount margin, $\delta = r - \mu_c$
Q	Annual captured emissions (t/yr), the annual CO ₂ output multiplied by the capture rate
$\beta_{\text{lo}}, \beta_{\text{hi}}$	Positive roots of the characteristic equation at the low-volatility and high-volatility endpoints of the declared ambiguity set
β_1	Positive root of the characteristic equation, Eq. (3), as a function of σ_p
β_0	Shorthand for $\beta_1(\bar{\sigma}_0)$, the root at the volatility anchor
Δ_β	Spread between the roots at the two volatility endpoints, $\Delta_\beta := \beta_{\text{lo}} - \beta_{\text{hi}} \geq 0$, strictly positive at a positive radius (Proposition 3)
D_0, E_0	Auxiliary curvature terms in the closed form of $\tilde{\kappa}$, Eq. (7)

Ambiguity-radius and recorded-weight parameters

α	Recorded weight on the adverse (low-volatility) endpoint of Σ_t under the α -MEU parameterization. It is elicited or configured before the gate and serves as a reporting input. At $\alpha = 1$, the benchmark objective reduces to MEU
Σ_t	Volatility ambiguity set, Eq. (1)
h_t	Admissible ambiguity radius at gate t ; in the application, $h_t = \min\{c_w w_t, (1 - \epsilon)\bar{\sigma}_0\}$
c_w	Signal-to-radius scaling constant in the pre-cap map $h_t = c_w w_t$, $c_w \geq 0$
ϵ	Admissibility margin for the ambiguity-radius cap, $0 < \epsilon < 1$
ρ	Relative ambiguity radius, $h/\bar{\sigma}_0$. Variants with an accent, a qualifier, or a subscript denote other quantities, namely $\hat{\rho}_k$ (sample autocorrelation), Spearman ρ (rank correlation), and ρ_{retest} (reliability)
$\tilde{\alpha}(h)$	Critical weight at which the member trigger crosses the anchor trigger, for $h > 0$, Eq. (6); $\tilde{\alpha}(0) := 1/2$ by continuous extension
$\tilde{\alpha}_{\text{lin}}, \tilde{\alpha}_{\text{rec}}, \tilde{\alpha}_{\alpha\text{-MEU}}$	Critical weight under the affine rule, the recursive rule, and the pre-commitment benchmark
$\sigma_{\text{eff}}(\alpha)$	Effective volatility of the recursive rule, $\sqrt{\alpha\sigma_{\text{low}}^2 + (1 - \alpha)\sigma_{\text{high}}^2}$
$\alpha_{\text{min}}, \alpha_{\text{max}}$	Smallest and largest recorded weights on a panel, which set the two interval edges
$\tilde{\kappa}(\bar{\sigma}_0)$	Volatility-channel curvature ratio, $V^{*''}/(4V^{*'})$, Eq. (7)

Approximation and safeguard quantities

θ	Registered economic-calibration tuple $(r, \mu_c, \bar{\sigma}_0)$
$e_{\mathcal{R}}(\theta, \rho)$	Registered refined-grid maximum of the anchor-normalized error between affine and benchmark member triggers, Eq. (18)
$E_{\mathcal{R}}(\theta, \rho)$	Price-unit release envelope specific to a calibration and a radius, Eq. (18)
$d_{\text{edge}}, d_{\text{member}}$	Price-edge and nearest-member distances used by the safeguard, Eq. (18)
ϵ_{num}	Numerical tolerance added to the registered maximum error

Policy-text input quantities

w_t	Quarterly rolling variability signal, the four-quarter population standard deviation of PSDI, Eq. (20)
PSDI _{t}	Policy-semantic-dispersion index, Eq. (19)
$\tilde{c}_t$	Centroid of normalized document embeddings in period t

Abbreviations

α -MEU	Alpha-maxmin expected utility
GBM	Geometric Brownian motion
MEU	Maxmin expected utility
PSDI	Policy-semantic-dispersion index

Notes. Symbols appearing in at least one numbered equation or at least six times in the mathematics are listed.

construction into asset pricing. Formulations concerned with how an agent should act return a policy. Matoussi et al. (2015) solved robust utility maximization in nondominated models through second-order backward stochastic differential equations. Källblad et al. (2018) obtained dynamically consistent investment criteria under model uncertainty through robust forward preferences. Formulations concerned with stopping, such as multiple-prior and time-consistent treatments, return a stopping region or an equilibrium policy (Riedel, 2009; Huang and Yu, 2021). Formulations concerned with commitment return a single robust decision. Recursive multiple priors belong to this group (Chen and Epstein, 2002; Epstein and Schneider, 2003). So do robust control (Hansen and Sargent, 2001) and distributionally robust optimization (Mohajerin Esfahani and Kuhn, 2018). They evaluate uncertainty at the level of the decision problem and return one decision robust over the whole set. A robust recursion returns the same type of object, and a rectangularity condition preserves its Bellman structure (Iyengar, 2005). The ambiguity set may also be a confidence region estimated from observed data (Wiesemann et al., 2013), and the reported output remains one worst-case policy. Under smooth ambiguity (Klibanoff et al., 2005), Mazzon and Tankov (2024) reduced a coal-plant divestment problem with learning to a family of standard stopping problems. Their output is still the policy of a single decision maker. Values, policies, regions, and robust decisions are all indexed by the problem, so none of the reviewed formulations returns the price range of member disagreement.

One strand of the ambiguity literature works with sets of valuations and comes closer to such a range. In the Knightian decision theory of Bewley (2002), incomplete preferences represent ambiguity. One act ranks above another only if its expected utility is higher under every prior in the set. Some pairs therefore remain unranked, and an inertia assumption resolves them in favor of the status quo. An alternative is then adopted only when it dominates the status quo unanimously. Rigotti and Shannon (2005) carried this structure into general equilibrium where incomplete preferences induce a set of valuations. The unanimity principle of Bewley (2002) suggests an analogous reporting logic for stage-gate screening, because deferral is the incumbent option at a gate. A unanimous invest classification here records agreement among the recorded member rules at the prevailing price. Where unanimity fails, aggregation is one possible downstream step. Axiomatic treatments of this step exist for heterogeneous beliefs and ambiguity attitudes. Gilboa et al. (2004) restricted the Pareto principle to choices under identical beliefs. Crès et al. (2011) aggregated maxmin experts sharing a utility function. Danan et al. (2016) applied the Pareto criterion only to comparisons robust to belief imprecision. Each of these studies characterized a social criterion, whereas the member-threshold vector precedes the aggregation step.

The contested-price interval is computed from the member-threshold vector, whose entries are indexed by recorded weights. A project-level formulation can supply such a vector only after an explicit member-indexing step. For the pre-commitment objective, this step solves the same scalar problem at each recorded weight. A time-consistent formulation yields scalar thresholds only if the equilibrium stopping region of every member has a

unique boundary in the same ordered state variable. The ambiguity set used here meets this condition.

The applied literature on CCUS retrofit timing also works at the project level. Compennolle et al. (2017) evaluated offshore enhanced oil recovery with CO₂ storage. Fan et al. (2023) and Li et al. (2026) examined the retrofit timing of coal-fired plants, and Sun et al. (2024) examined incentive design. Colombe et al. (2024) worked at the system level and derived a deployment plan through two-stage stochastic programming. Deeney et al. (2021) built a real-options decision support tool for research and development investment in CO₂ recycling. Their compound option structure carries technical and market risk and values the project.

By contrast, one line of theory indexes the option by the individual. Garlappi et al. (2017, 2022) examined group real options in which members hold heterogeneous beliefs and decide by voting. In their model, a representative median member cannot reproduce group behavior. Their analysis concerns group behavior after a vote, whereas the present report precedes any vote. The present setting differs from this literature in three further respects. Heterogeneity enters through ambiguity attitudes, all members share the same information set, and ambiguity falls on volatility. Cardenas (2025) extended this line to the timing of group investment decisions. In this model, members hold heterogeneous beliefs, and the rule of each member has a double continuation region. Disagreement can produce inertia and underinvestment even when every member would invest individually. The two boundaries delimit the continuation region of one member, whereas the interval reported here spans the one-sided thresholds of different members. The gate reports this interval without resolving the disagreement. Across this stream, the reported objects are project-level triggers, values, policies, stopping regions, robust decisions, and group behavior after a vote. In the reviewed work, member-level rules appear inside group models of voting or joint timing (Garlappi et al., 2017; Cardenas, 2025). The first gap is therefore the absence of a gate-level vector of one-sided member thresholds indexed by ambiguity attitude. This gap includes the position of such thresholds relative to the no-ambiguity anchor under volatility ambiguity.

2.2. Stage-gate screening and committee-level decision support

Stage-gate processes organize investment evaluation as sequential committee reviews (Cooper, 2008). The supporting decision systems typically deliver aggregate scores, rankings, or recommendations (Sprague, 1980; Power, 2002), which are updated as conditions change (Violos et al., 2025). Group decision methods such as TOPSIS-based scoring (Hwang and Yoon, 1981) also compress heterogeneous evaluations into a single score or ranking. A score ranks alternatives at one decision point, whereas a member-specific threshold is the state-variable level at which the screening classification of a member changes. A committee can use both, with the thresholds locating the prevailing price relative to each member and a scoring procedure ranking the remaining alternatives. The interval width measures dispersion among model-implied screening thresholds in price

units. By contrast, consensus measures, reviewed by Herrera Viedma et al. (2014), assess alignment among stated preferences before a collective choice.

A separate operational-research tradition treats disagreement as an output of the analysis. Phillips (1984) built the decision model jointly with the problem owners and counted it as required site once further sensitivity analysis generated no new intuitions. Franco and Montibeller (2010) developed this stance into facilitated modeling for groups. The present method offers this tradition a quantitative device for one setting, translating recorded weights into thresholds without a further aggregation step. Two groups of recent methods operate downstream of the resulting report: aggregation procedures and procedures deciding above the individual member. Dong et al. (2026) integrated information across attributes and periods to produce superiority-probability comparisons and probabilistic rankings. Song and Li (2026) aggregated numerical and linguistic preference relations at the subgroup level and then solved minimum-adjustment consensus models under limited compromise. Portfolio and multi-stage models kept the decision above the individual member (Marques et al., 2022; Shen and Li, 2023; Saiz et al., 2024). Recent robust and adaptive designs did so as well (Shen et al., 2025; Shavazipour and Stewart, 2026). They use an uncertainty set to select one policy with acceptable performance across the set. These studies address what the organization should commit to, whereas the present method examines whether the members screening one project currently agree. For each member, it reports the signed margin, defined as the member threshold minus the prevailing price. The margin is positive under a wait classification. At the committee level, the method reports the interval edges. Aggregating the edges away at the gate would leave a single number without a record of the member behind it. A later consensus model, ranking, or portfolio selection could use the member thresholds, given a stated rule for how they enter its objectives and constraints. Most reviewed methods aggregate or reconcile member assessments before reporting, and facilitated modeling lacks a quantitative threshold device. No reviewed method converts recorded member weights into thresholds on the price axis, and this missing report is the second gap.

2.3. Text-based uncertainty measures as a model input

In the application, only one gate input, the ambiguity radius, comes from policy text. Existing studies commonly derive text-based uncertainty measures from keyword frequencies or supervised classifiers. The Economic Policy Uncertainty index counts prespecified newspaper terms (Baker et al., 2016). Investment research often uses it as a regression covariate (Gulen and Ion, 2016). Hassan et al. (2019) extended this design to firm-level political risk, and Gavrilidis (2021) applied it to climate news. Fixed term lists may miss paraphrases and emerging terminology. Hartley (2025) reported closer agreement with human-audited labels for large language model classifiers than for dictionary rules. Whether policy documents carry a usable dispersion signal remained open in this comparison. Supervised alternatives, such as the MacBERT-based climate policy uncertainty index for China of Ma et al. (2023), require labeled training data. The limits of fixed term lists and labeled data

motivate the unsupervised, embedding-based construction in the application. This construction follows the text-as-data approach reviewed by Gentzkow et al. (2019) and yields a series measuring semantic dispersion across contemporaneous policy documents. In the reviewed studies, text-based measures served as indices, document labels, or regression covariates. None of them supplied an input to a member-level screening rule, which is the third gap.

Table 1 summarizes the three gaps. First, the reviewed real-options formulations return project-level objects or group behavior, and none reports a gate-level vector of member thresholds indexed by ambiguity attitude. Second, committee decision support aggregates member assessments into scores, rankings, or consensus measures, so the price range of member disagreement stays unreported. Third, text-based uncertainty measures have not been linked to member-level screening rules. The present method addresses the first two gaps with a member-indexed threshold vector on the price axis. The critical weight locates this vector relative to the anchor, and a stated accuracy standard sets its release conditions. For the third gap, the application supplies a replaceable text-based radius as one gate input.

3. Decision setting and model inputs

The member-threshold vector addressing the first two gaps is built from the gate decision and its inputs. These inputs are of two kinds: inputs shared by all members and one recorded weight per member. Each reported threshold combines the shared model inputs with the weight of one member.

3.1. The gate-level decision problem

At each gate of a stage-gate sequence, the project receives one invest-or-wait classification per member before the committee makes its collective decision. The committee also declares an interval of forward volatilities, the ambiguity set of the gate, which Eq. (1) parameterizes by one radius around an anchor. Members share one regulatory information set and differ only in the weight $\alpha_i \in [0, 1]$ recorded for member i on the adverse endpoint.

The shared inputs are a calibration $\theta = (r, \mu_c, \bar{\sigma}_0)$, the project inputs I and Q , the declared ambiguity radius h_t , and the prevailing price P_t . The member-specific inputs are the recorded weights $\{\alpha_i\}_{i=1}^m$. From these inputs, the method returns the member-specific price thresholds and classifies the gate at P_t as unanimous wait, mixed, or unanimous invest. It also states whether the affine report may be released alone or requires a benchmark recomputation.

Figure 1 organizes these inputs and outputs into an input layer, an analytical layer, and an interface layer. The interface layer applies the release rule and issues either an affine-only report or a dual report containing both the affine and the benchmark results.

3.2. Model inputs and assumptions

The analysis rests on three maintained model assumptions, one reporting convention, and one application-side input. (i)

Table 1. Relation of prior literature streams to the contested-price interval and its width. The first two rows are the streams underlying the method, and the third row covers the application-specific input of the running example.

Literature stream	Contribution	Gap for stage-gate screening	Role in this paper
Ambiguity-aware real options (Nishimura and Ozaki, 2007; Trojanowska and Kort, 2010; Schröder, 2011; Fan et al., 2023)	Investment timing rules under non-unique priors	Gate-level vector of member thresholds indexed by ambiguity attitude	Analytical basis for member triggers
Decision support and stage-gate screening (Sprague, 1980; Cooper, 2008; Shen and Li, 2023; Saiz et al., 2024; Shen et al., 2025; Violos et al., 2025)	Sequential review processes and aggregate scores or rankings	Explicit reporting of member-threshold dispersion from recorded weights	Interface logic reporting the contested-price interval, its width, and the current classification summary
Application input. Text-based policy-uncertainty measurement (Baker et al., 2016; Gavrilidis, 2021; Ma et al., 2023; Hartley, 2025)	Quantitative signals from policy and news text	Integration of such signals into member-specific gate-level screening rules	Construction of the replaceable input w_t

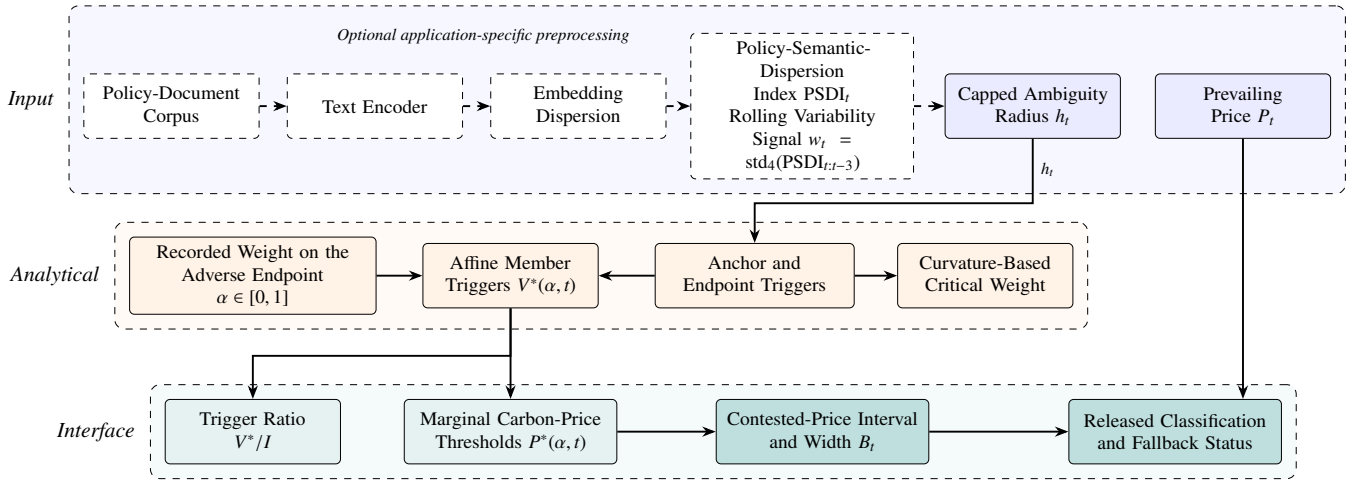


Figure 1. Architecture of the screening method in three layers. The input layer supplies the capped ambiguity radius h_t and the prevailing price P_t . Of these two inputs, only h_t enters the member thresholds and the interval, whereas P_t enters at the classification step. The dashed boxes show the application-specific derivation of h_t , from a policy-document corpus through the dispersion index $PSDI_t$ to the rolling variability signal w_t . A committee declaring its radius directly omits these steps. The analytical layer maps each recorded weight on the adverse endpoint to a member trigger and locates the curvature-based critical weight. The interface layer converts the triggers into trigger-to-cost ratios and marginal carbon-price thresholds. It reports the contested-price interval, its width B_t , the released classification, and the release or fallback status.

Forward volatility carries the ambiguity, and drift is held fixed (ii) Standard deviation parameterizes the ambiguity set. (iii) The carbon-price process follows a GBM with $r > \mu_c > 0$. The reporting convention is the affine endpoint rule, which defines the member triggers, and the application-side input is the ambiguity radius. Within a gate, the common information set stays fixed, and so do the recorded weights, which are reporting inputs registered in advance. Each member then receives a separate invest-or-wait classification.

Under assumption (i), ambiguity concerns the width of the forward distribution and leaves its mean unaffected. The policy-text signal of the application fits this reading, because it records variation in semantic dispersion and carries no information about the direction of policy. The signal therefore configures the ambiguity radius, and the growth scenario μ_c stays fixed. The available record left the scale of the signal-to-radius map unsettled, so this map is treated as a convention. Forward volatility enters as a single parameter, and assumption (ii) declares its ambiguity set in standard-deviation units. This declaration fixes the coordinate for evaluating trigger curvature and, with it, the

sign of the critical-weight displacement.

The ambiguity radius enters the model as a single number per gate. The forward-volatility anchor $\bar{\sigma}_0$ and an admissibility margin $\epsilon \in (0, 1)$ fix its admissible range. The anchor stays fixed across gates within an application. The radius of each gate is capped at $(1 - \epsilon)\bar{\sigma}_0$, which keeps the lower volatility endpoint strictly positive. The ambiguity set is

$$\sigma_p \in \Sigma_t = [\bar{\sigma}_0 - h_t, \bar{\sigma}_0 + h_t], \quad (1)$$

where σ_p denotes forward volatility. A committee may declare the gate radius h_t directly or derive it from an auxiliary series, as in the application. Radii quoted as percentages are relative radii $h_t/\bar{\sigma}_0$. The primary radius range covers relative radii up to 22%, and the extended range adds a 33% stress radius. Every numerical result is evaluated on a finite grid, and its scope is stated with the result.

These definitions complete the inputs of the gate. The remaining task is analytical: mapping each weight to a trigger, converting the triggers into price thresholds, and stating when the report may be released.

4. Model

The analytical task starts from the single trigger of a conventional real-options analysis. The model extends this trigger to one trigger per recorded weight and attaches a release rule to the resulting member-threshold vector.

4.1. Baseline screening trigger

For a given $\sigma_p \in \Sigma_t$, the baseline trigger of McDonald and Siegel (1986) is

$$V^*(\sigma_p) = \frac{\beta_1}{\beta_1 - 1} I, \quad (2)$$

where I is the investment cost. The exponent $\beta_1 > 1$ is the positive root of the characteristic equation

$$\frac{1}{2}\sigma_p^2\beta(\beta - 1) + \mu_c\beta - r = 0. \quad (3)$$

Here μ_c is the annual carbon-price growth scenario declared by the committee as a decision input, and r is the discount rate. Assumption (iii) requires $r > \mu_c > 0$, which guarantees $\beta_1 > 1$ and a finite trigger at every positive volatility (Dixit and Pindyck, 1994). This condition defines the admissible parameter domain, and for a calibration outside it, the committee records the absence of a finite trigger. The trigger in Eq. (2) is strictly increasing in σ_p (Dixit and Pindyck, 1994). Its minimum and maximum over the ambiguity set therefore lie at the endpoint volatilities $\sigma_{\text{low}} := \bar{\sigma}_0 - h_t$ and $\sigma_{\text{high}} := \bar{\sigma}_0 + h_t$:

$$V_{\text{low}}^* = V^*(\sigma_{\text{low}}), \quad V_{\text{high}}^* = V^*(\sigma_{\text{high}}). \quad (4)$$

Proposition 1 (Strict convexity). *For all $r > \mu_c > 0$, the investment trigger $V^*(\sigma_p)$ is strictly convex in σ_p on the ambiguity set Σ_t .*

The Supplementary Material contains the proofs of all propositions and corollaries. The proof of Proposition 1 writes $V^{*''}(\sigma_p)$ as a positive prefactor times the bracket $\sigma_p^4(4\beta_1 - 3) + 4\mu_c\sigma_p^2 + 4\mu_c^2$, whose three terms are positive when $\beta_1 > 1$ and $\mu_c > 0$.

Define the anchor trigger $V_0^* := V^*(\bar{\sigma}_0)$. At any positive radius h_t , strict convexity makes the upper endpoint gap exceed the lower one, $V_{\text{high}}^* - V_0^* > V_0^* - V_{\text{low}}^*$. When both gaps are expanded about the anchor volatility, their odd-order terms cancel in the difference, which therefore equals $V^{*''}(\bar{\sigma}_0)h_t^2 + O(h_t^4)$. This asymmetry displaces the critical weight away from one-half.

4.2. Member-specific screening triggers

The baseline trigger is common to all members. Member heterogeneity enters through a recorded weight $\alpha \in [0, 1]$ on the adverse endpoint, written in the α -MEU parameterization (Ghirardato et al., 2004). The recorded weight α_i is the response of member i to an elicitation instrument, mapped onto $[0, 1]$ by a stated normalization. Here the adverse endpoint is the low-volatility endpoint, because lower volatility reduces the value of the option to wait (Dixit and Pindyck, 1994). More weight on this endpoint therefore lowers the member trigger and the corresponding member threshold. The affine endpoint rule

defines the member trigger as a convex combination of the two endpoint triggers:

$$V^*(\alpha, t) = \alpha V_{\text{low}}^* + (1 - \alpha) V_{\text{high}}^*. \quad (5)$$

The two endpoint triggers carry the time subscript through h_t . The ordering of member triggers follows from $V_{\text{low}}^* < V_{\text{high}}^*$ alone, and the pre-commitment α -MEU benchmark yields the same ordering at every admissible calibration.

Three properties motivate the affine choice. First, it preserves a monotone correspondence between recorded weights and member thresholds, which keeps the member-threshold vector interpretable. Second, it has a closed form, which keeps the critical-weight derivation transparent. Third, it reproduces both endpoint triggers without an arbitrary reference project value, which the pre-commitment benchmark requires. A one-parameter nonlinear family of rules reduced the trigger error against the benchmark, yet it lost the above-one-half location of the critical weight. It also needed a radius-specific fit to align the directional labels and added one item to the registered configuration.

The recorded weight is not inferred from behavior, because observed choices alone need not identify a unique interior α (Eichberger et al., 2011). This paper treats α as a reporting input, elicited or configured institutionally. A smooth-ambiguity member map would index each member by a second-order belief over priors and an ambiguity-aversion function (Klibanoff et al., 2005). Chen et al. (2025) derived optimal payoffs under these preferences in a one-period market. The present design, by contrast, needs one recorded weight per member.

Given the recorded weights, the affine rule also supports a directional reading of each member trigger. At a positive radius, a member trigger below the anchor trigger is acceleration-oriented, and one above it is delay-oriented. These directional labels complement the invest-or-wait classifications. The direction of these labels can be checked against the drift-ambiguity result of Nishimura and Ozaki (2007). At $\alpha = 1$ and a positive radius, the shift under volatility ambiguity runs in the same qualitative direction as their maxmin result.

4.3. Curvature-based critical-weight mechanism

The boundary between the two directional labels is a single weight. To derive it, let h denote a generic positive admissible value of h_t . Under the affine rule, the member trigger is strictly decreasing in the recorded weight. At a positive radius, the anchor trigger lies strictly between the two endpoint triggers. Exactly one weight, called the critical weight, therefore places the member trigger at the anchor trigger. The two endpoint gaps determine its location, and it equals one half only if these gaps are equal. The curvature ratio $\tilde{\kappa}$ in Eq. (6) gives its leading-order displacement from one half as the radius approaches zero. Corollary 1 establishes a positive displacement at every admissible radius.

Proposition 2 (Critical weight, exact expression and local expansion). *For $0 < h \leq (1 - \epsilon)\bar{\sigma}_0$, define the ambiguity premium*

as $AP(\alpha) := V^*(\alpha, t) - V_0^*$. The premium has a unique zero

$$\tilde{\alpha}(h) = \frac{V_{\text{high}}^* - V_0^*}{V_{\text{high}}^* - V_{\text{low}}^*} = \frac{1}{2} + \tilde{\kappa}(\bar{\sigma}_0)h + O(h^2),$$

$$\tilde{\kappa}(\bar{\sigma}_0) := \frac{V^{*''}(\bar{\sigma}_0)}{4V^{*'}(\bar{\sigma}_0)} > 0.$$

Primes denote derivatives of $V^*(\sigma_p)$ with respect to σ_p at the anchor volatility. The expansion follows from the two endpoint gaps. Their sum forms the denominator, in which the $O(h)$ terms add and the $O(h^2)$ terms cancel. The numerator $V_{\text{high}}^* - V_0^*$ is the upper gap, in which the curvature term survives. To first order, the quotient is therefore $1/2 + \tilde{\kappa}(\bar{\sigma}_0)h$. At $h = 0$, $AP(\alpha) = 0$ for every $\alpha \in [0, 1]$, and no unique critical weight exists, so the continuous extension $\tilde{\alpha}(0) := 1/2$ is used for reporting. All numerical results use the exact expression in Eq. (6), whereas the local expansion interprets behavior near $h = 0$.

The curvature ratio $\tilde{\kappa}$ has a closed form:

$$\tilde{\kappa}(\bar{\sigma}_0) = \frac{1}{4\bar{\sigma}_0} - \frac{\bar{\sigma}_0(\beta_0 - 1)E_0}{D_0^2},$$

where $\beta_0 := \beta_1(\bar{\sigma}_0)$, $D_0 := \bar{\sigma}_0^2(2\beta_0 - 1) + 2\mu_c$, and $E_0 := \bar{\sigma}_0^2(\beta_0 - 1) + 2\mu_c$. The first term of Eq. (7) comes from reparameterizing variance as standard deviation and pushes the critical weight above one half. The second term reflects concavity of the trigger in variance and pushes in the opposite direction. The first term dominates on the admissible domain $r > \mu_c > 0$. This dominance is the algebraic counterpart of Proposition 1 and makes $\tilde{\kappa}$ strictly positive. The same derivation establishes strict concavity of the trigger in σ_p^2 .

Corollary 1 (Global location of the critical weight). *For all $0 < h \leq (1 - \epsilon)\bar{\sigma}_0$, the exact critical weight satisfies $\tilde{\alpha}(h) > 1/2$*

Corollary 2 (Monotonicity of the critical weight). *The continuous extension of the positive-radius critical weight, defined by $\tilde{\alpha}(0) := 1/2$, is strictly increasing for $0 \leq h \leq (1 - \epsilon)\bar{\sigma}_0$, with right derivative $\tilde{\alpha}'(0+) = \tilde{\kappa}(\bar{\sigma}_0) > 0$.*

A recorded weight below the critical weight gives a positive ambiguity premium, and a weight above it gives a negative premium. A single condition on the trigger, stated independently of the present model, locates this sign change relative to one half. Take any increasing scalar trigger $G(x)$ at the symmetric ordered endpoints $x_0 - h$ and $x_0 + h$, with the adverse weight on the lower endpoint. The affine critical weight exceeds one half if and only if the midpoint Jensen gap $G(x_0 + h) + G(x_0 - h) - 2G(x_0)$ is positive. Locally, the displacement of the critical weight from one half has the sign of $G''(x_0)/G'(x_0)$. Corollary 1 therefore rests on convexity and the affine form alone. A vanishing Jensen gap places the critical weight exactly at one half.

Because the trigger is strictly concave in σ_p^2 , a declaration of the same relative radius in variance units reverses the Jensen gap. The critical weight then falls below one half, so its location depends on the coordinate of the declared ambiguity set. Figure 2 plots V^*/I against α under both declarations. Each curve crosses the anchor level V_0^*/I at its own critical weight, and the spread

of the ratio across weights widens with the radius. At a 10% relative radius, the critical weight was 0.517 in standard-deviation units and 0.496 in variance units, and the difference grew with the radius. At a matched relative radius, the two coordinates select different pairs of endpoint volatilities and therefore describe two different ambiguity sets. Assumption (ii) fixes the declared coordinate at standard deviation. The committee declares the ambiguity set in the units of its scenarios, and the directional labels are read against this declaration.

At a fixed calibration and radius, the affine label depends on the coordinate alone, whereas the benchmark label also depends on the reference project value.

Corollary 1 requires the Jensen-gap criterion, which presupposes one scalar coordinate with two ordered endpoints. Joint ambiguity over several channels lacks a natural pair of such endpoints, so the criterion needs a declared path or ordering and a rederived trigger.

4.4. Price relevance of the contested-price interval

The directional labels compare each member trigger with the anchor trigger, whereas the invest-or-wait classification is read on the price axis. Converting each member trigger $V^*(\alpha_i, t)$ to the carbon-price scale gives the marginal carbon-price threshold

$$P_i^* = \frac{V^*(\alpha_i, t)\delta}{Q}, \quad \delta = r - \mu_c, \quad (8)$$

where δ is the net discount margin. The annual captured emissions Q equal the annual CO_2 output of the plant multiplied by the capture rate assumed by Sun et al. (2024). Eq. (8) equates the trigger with the present value P^*Q/δ of perpetual carbon-price-linked cash flows. The no-ambiguity threshold $P_0^* := V_0^*\delta/Q$ is called the anchor threshold. Eq. (8) multiplies every member trigger by the same positive factor, so the thresholds inherit the critical weight defined on triggers. Define the contested-price interval and its width as

$$C_t := \left[\min_i P_i^*, \max_i P_i^* \right],$$

$$B_t := \text{width}(C_t) = \max_i P_i^* - \min_i P_i^*.$$

The recorded weight vector $\{\alpha_i\}$ is the weight profile of the gate. For $h_t > 0$, $P^*(\alpha)$ is affine and strictly decreasing, so two profiles with the same extreme weights yield the same interval at a fixed radius. Because the difference between the two endpoint triggers is $2V^{*'}(\bar{\sigma}_0)h_t + O(h_t^2)$, the width factorizes to leading order as

$$B_t \approx 2(\alpha_{\max} - \alpha_{\min})V^{*'}(\bar{\sigma}_0)h_t \frac{\delta}{Q}.$$

The width is thus approximately the product of the recorded-weight range, the ambiguity radius, and the local trigger sensitivity in price units, $2V^{*'}(\bar{\sigma}_0)\delta/Q$. This approximation reproduced the exact width within 0.06% over the primary radius range. At $h_t = 0$, the width is zero, and no price produces a mixed classification. The price-relevance condition uses the full min-max interval of Eq. (9).

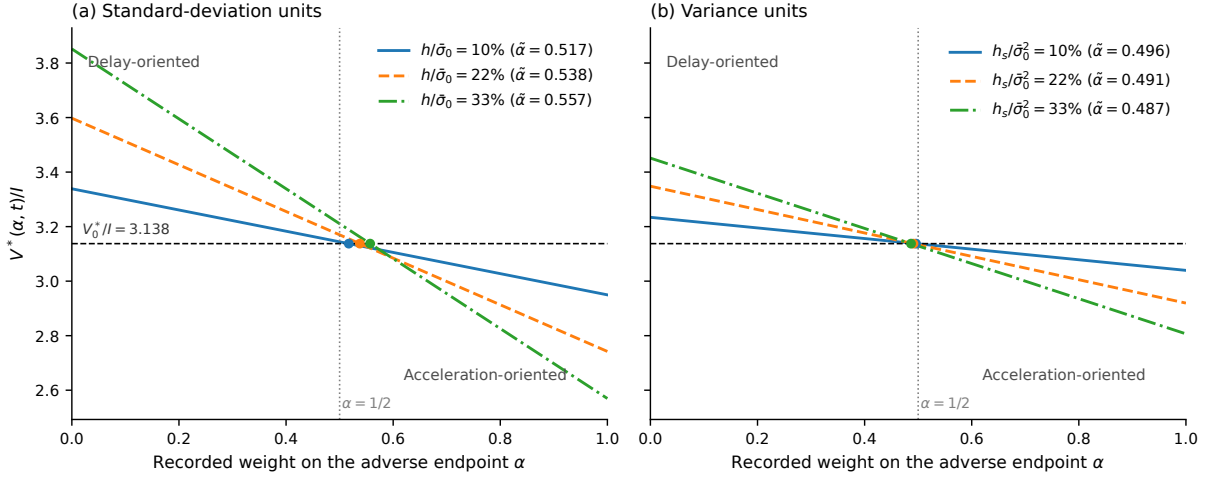


Figure 2. Trigger ratio V^*/I across recorded weights at three ambiguity radii. Directional labels indicate shifts from the no-ambiguity anchor. Panel (a) declares the ambiguity set in standard-deviation units, and panel (b) declares the same relative radius in variance units. Markers show the critical weight of each curve. The trigger is convex in σ_p and concave in σ_p^2 . The critical weight therefore lies above one half in panel (a) and below one half in panel (b).

Price-relevance condition. At a gate, let P_t denote the prevailing effective carbon price, called the prevailing price. The classification rule assigns member i an invest classification when $P_t \geq P_i^*$ and a wait classification otherwise. Both classifications occur among members if and only if

$$P_t \in C_t \iff \min_i P_i^* \leq P_t < \max_i P_i^*. \quad (11)$$

Outside C_t , all members receive the same classification, even when their thresholds differ.

Basis of the threshold. The comparison between the prevailing price and the member thresholds is meaningful only on a stated threshold basis. Under the stated cash-flow mapping, P^* is a marginal threshold, which excludes operating costs, the energy penalty, transport and storage costs, and subsidies. Excluded costs belong to the project, so an excluded cost expressible as a common per-tonne adjustment shifts every member threshold equally.

Common adjustments. Two adjustments of practical interest act identically on every member threshold, one additively and one multiplicatively. A common per-tonne cost or subsidy shifts all thresholds by the same amount. With the perpetual trigger ratios held fixed, replacing the perpetuity with a common finite-horizon annuity multiplies all thresholds by the same factor. Both adjustments are monotone and preserve the equality between a member threshold and the anchor threshold. The member ordering and the critical weight therefore stay unchanged, while the classification at a fixed prevailing price may move. The additive shift leaves the interval width and the absolute threshold error of the affine rule unchanged, whereas the multiplicative factor rescales both. Under the factor, the error therefore stays unchanged only in relative terms.

Let s denote a common per-tonne subsidy and c a common per-tonne cost, and define $\Delta = c - s$. Every member threshold

becomes $P_{i,obs}^* = P_i^* + \Delta$, and the interval $C_t = [L, U)$ becomes $[L + \Delta, U + \Delta)$. A state-dependent or quantity-dependent component requires a different project-value specification.

Member-specific adjustments. An adjustment varying across members changes the interval width by at most the spread of the adjustments. It can also reverse the member ordering and break the single equality defining the critical weight.

4.5. Trigger formulations and the accuracy of the affine rule

The affine rule is a reporting convention, so its member triggers are compared with triggers derived from an explicit decision criterion. Table 2 lists the affine rule and three alternative formulations, and all except the equilibrium of Huang and Yu (2021) are evaluated numerically. The pre-commitment α -MEU optimum serves as the accuracy standard and is compared with the affine rule at the same granularity, one trigger per member. The recursive α -mixture, which applies the same weight at every state, serves as an analytical comparator. Here, $V_{lin}^*(\alpha)$ denotes the member trigger of Eq. (5), and $V_{\alpha-MEU}^*(\alpha, V_0)$ denotes the benchmark optimum at the same recorded weight.

4.5.1. The pre-commitment benchmark

For a trigger V^* and a reference project value $V_0 < V^*$, let τ denote the first hitting time of V^* . Under a constant volatility σ , the expected discounted payoff at τ takes the standard real-options form

$$\mathbb{E}_\sigma[e^{-r\tau}(V_\tau - I)] = \left(\frac{V_0}{V^*}\right)^{\beta_1(\sigma)}(V^* - I). \quad (12)$$

In the benchmark, the ambiguity set is represented by the pair of constant-volatility measures indexed by the declared endpoints. The factor $(V_0/V^*)^{\beta_1}$ decreases in β_1 when $V_0 < V^*$, and $\beta_1(\sigma)$ decreases in σ . The infimum and supremum of Eq. (12) over the set are therefore attained at σ_{low} and σ_{high} , respectively. No

admissible calibration reverses this endpoint ordering. A reversal would require a reference value above the trigger, or an exercise payoff nonlinear enough to make the project value decrease in volatility. Placing weight α on the infimum and $1 - \alpha$ on the supremum gives the α -MEU objective of a trigger rule

$$J^\alpha(V^*, V_0) = (V^* - I) \times \left[\alpha \left(\frac{V_0}{V^*} \right)^{\beta_1(\sigma_{\text{low}})} + (1 - \alpha) \left(\frac{V_0}{V^*} \right)^{\beta_1(\sigma_{\text{high}})} \right]. \quad (13)$$

The optimal trigger $V_{\alpha\text{-MEU}}^*(\alpha, V_0) \equiv \arg \max_{V^*} J^\alpha(V^*, V_0)$ has no closed form for $\alpha \in (0, 1)$, so bounded scalar optimization computes it. Solving this scalar problem at each recorded weight yields the benchmark member triggers, and Eq. (8) converts them into the benchmark member-threshold vector. Because the maximization runs over constant triggers, the solution is optimal within the class of constant-trigger rules under the pre-commitment criterion.

The benchmark vector depends on the reference project value V_0 , because the pre-commitment objective is evaluated directly at this value without a recursive value function. The benchmark is defined for V_0 below the benchmark trigger of every member, a region called the benchmark domain. We fix $V_0 = I$, the value at which immediate investment just breaks even. A calibration-radius cell entered the evaluation only when V_0 lay below the low-endpoint trigger V_{low}^* . By Proposition 3(i), every stationary point of J^α lies in $[V_{\text{low}}^*, V_{\text{high}}^*]$, so this entry rule is a conservative sufficient form of the domain condition. As a result, $V_0 = I$ lay in the benchmark domain at every evaluated cell. The affine and benchmark calculations use the same economic primitives, volatility endpoints, recorded weights, and radius grid.

The benchmark vector qualifies as the accuracy standard under two conditions. The computed maximizer must be global and its position relative to the affine trigger must be known. Proposition 3 settles both through one quantity, the spread between the two endpoint exponents.

Proposition 3 (Benchmark well-posedness and one-sidedness). Write $\beta_{\text{lo}} := \beta_1(\sigma_{\text{low}})$ and $\beta_{\text{hi}} := \beta_1(\sigma_{\text{high}})$ for the exponents at the two endpoints, and $\Delta_\beta := \beta_{\text{lo}} - \beta_{\text{hi}} > 0$ for their spread. Fix a recorded weight α and a reference project value V_0 in the benchmark domain.

(i) *Unique global maximizer.* Every stationary point of J^α lies in $[V_{\text{low}}^*, V_{\text{high}}^*]$. If

$$\Delta_\beta \leq 1 \quad \text{or} \quad (\Delta_\beta - 1) V_{\text{high}}^* \leq \Delta_\beta V_{\text{low}}^*, \quad (14)$$

the stationary point is unique and is the global maximizer. Condition (14) is free of α and of V_0 .

(ii) *One-sidedness.* Assume Condition (14). For $\alpha \in (0, 1)$,

$$V_{\text{lin}}^*(\alpha) \geq V_{\alpha\text{-MEU}}^*(\alpha, V_0) \iff \frac{\beta_{\text{lo}} - 1}{\beta_{\text{hi}} - 1} \geq \left(\frac{V_{\text{lin}}^*(\alpha)}{V_0} \right)^{\Delta_\beta}. \quad (15)$$

The affine and benchmark triggers coincide at the endpoint weights $\alpha = 0$ and $\alpha = 1$. Since V_{lin}^* decreases in α replacing it by its largest value V_{high}^* yields a condition sufficient for every recorded weight at once. At $V_0 = I$ this sufficient condition holds whenever $\Delta_\beta \leq 1$.

Conditions (14) and (15) have closed forms in the calibration and the radius, so a gate can check them without solving the benchmark. The economic calibrations form a 36-point Cartesian grid, and removing three cells violating $r > \mu_c$ leaves 33 admissible calibrations. Condition (14) held at all of them across the admissible radius range, with a slack of at least 0.9067. The global-maximizer property therefore held for every reported benchmark value. A separate scan counted exactly one interior maximum of J^α in each of 30,090 cells defined by calibration, radius, reference value, and weight. As a check on its detection power, the same counter flagged a deliberately bimodal control function. Condition (15) held at all 33 calibrations and every registered radius up to 22%. At the 33% stress radius, it failed at one calibration, with $r = 0.12$, $\mu_c = 0.02$, and $\bar{\sigma}_0 = 0.199$. There, the affine trigger fell below the benchmark for recorded weights near 0.01, by a relative gap of at most 4.5×10^{-6} confirmed at 40-digit precision. This failure lay above the 10% release cap, where a dual report is already mandatory.

4.5.2. The recursive alternative

The pre-commitment objective is evaluated once, at the gate, whereas applying the same α -mixture at every state yields a recursive rule, dynamically consistent by construction. In this setting, the rule also has a closed form. Under convexity of the continuation value, the α -mixture of the two endpoint generators collapses to the ordinary geometric Brownian generator at the effective variance

$$\sigma_{\text{eff}}^2(\alpha) = \alpha \sigma_{\text{low}}^2 + (1 - \alpha) \sigma_{\text{high}}^2. \quad (16)$$

The recursive trigger is then $V_{\text{rec}}^*(\alpha) = V^*(\sigma_{\text{eff}}(\alpha))$, which is free of any reference project value. Since V^* is strictly increasing in σ_p , solving $\sigma_{\text{eff}}(\tilde{\alpha}) = \bar{\sigma}_0$ gives the recursive critical weight

$$\tilde{\alpha}_{\text{rec}}(h) = \frac{1}{2} + \frac{h}{4\bar{\sigma}_0}, \quad (17)$$

which exceeds one half at every positive radius.

Convexity also makes the effective variance state-independent, so the recursive problem reduces to the ordinary stopping problem at this variance. Its stopping region is bounded by a single trigger, so a time-consistent formulation meets the requirement of one scalar threshold per member in this setting. The recursive rule serves as a comparator because it reads a recorded weight differently. Mixing the two generators mixes variances, so the weight indexes an effective volatility. Under the affine rule, by contrast, the weight indexes a position between the two endpoint triggers.

Eqs. (17) and (7) show where the critical weights of the two rules differ. The recursive displacement is $h/(4\bar{\sigma}_0)$, and its coefficient of h equals the first term of Eq. (7). The affine coefficient $\tilde{\alpha}$ also retains the negative second term, so near the anchor the affine critical weight lies below the recursive one. This ordering held at every evaluated radius in Table 4, which compares all three formulations. The affine and recursive critical weights lay above one half, whereas the pre-commitment one at $V_0 = I$ lay below it and decreased in the radius. Under

Table 2. Trigger formulations, their outputs, and their roles in this paper.

Formulation	Type of output	Dynamic consistency	Role in this paper
Affine endpoint rule, Eq. (5)	Member-trigger map $\alpha_i \mapsto V_i^*(t)$. Eqs. (8) and (9) convert it to $\{P_i^*\}$, C_i , and B_i	Not applicable (static gate-reporting map)	Reported rule in closed form, independent of the reference value, with one threshold per member
Pre-commitment α -MEU optimum, Eq. (13)	Member-trigger vector dependent on V_0 , evaluated at $\{\alpha_i\}$	No (pre-commitment rule)	Accuracy standard, recomputable at any gate, and required fallback
Recursive α -mixture of the generator, Eq. (16)	Member-trigger vector $V^*(\sigma_{\text{eff}}(\alpha_i))$, independent of V_0 , with a unique scalar boundary	Yes, by construction	Analytical comparator. Its critical weight has the closed form of Eq. (17) and lies above one half. The variance mixture fixes this weight for any increasing trigger map
Time-consistent α -MEU equilibrium (Huang and Yu, 2021)	Member-indexed equilibrium stopping regions. Scalar boundaries apply only when uniquely defined in a common state	Yes	Not evaluated. On the constant-endpoint set of this paper, the recursive row already gives a time-consistent rule with a single-threshold stopping region, as a threshold report requires. Under state-dependent prior sets, compressing the regions into $\{P_i^*\}$ is inadmissible

the benchmark at this reference value, both the sign of Corollary 1 and the monotonicity of Corollary 2 therefore reversed. The four-member reference profile has equally spaced recorded weights from 0.25 to 1.00. At the baseline calibration of Table 7 and this profile, the formulations disagreed far less on the interval width than on the directional labels. At a 10% radius, the widths were 10.945 CNY/t (affine), 11.027 CNY/t (recursive), and 10.717 CNY/t (pre-commitment). The two comparators lay within 2.1% of the affine width, and this distance grew with the radius to 6.4% at the 33% stress radius.

The closed forms in Eqs. (6) and (17) both rest on the perpetual geometric Brownian structure. Eq. (17) requires a convex continuation value, under which the α -mixture of the two endpoint generators collapses to Eq. (16). The displacement in Eq. (6) requires $V^*(\sigma_p)$ to be convex in σ_p . A different price process can break either property. The ratio in Eq. (6) was therefore recomputed under three further specifications: a finite investment horizon, a mean-reverting log price, and a jump diffusion. Each specification is solved on one grid shared by the anchor and the two endpoint volatilities. As a result, the discretization bias in the trigger level largely cancels in the ratio. Table 3 reports the smallest critical weight over each auxiliary grid and radius. The critical weight stayed above one half in all 52 evaluated cells with a smallest value of 0.5080. At a 25-year option horizon it differed from its perpetual value by at most 0.0020. Under jumps, the continuation value is no longer a power function of V and the GBM closed form for Eq. (6) is unavailable. The ratio in Eq. (6) still defines the critical weight when the specification yields two ordered scalar endpoint triggers. The trigger must also be monotone and curved in the declared coordinate. Under these two conditions, the critical weight is recomputed from the new continuation value as the weight equating the affine trigger with the anchor trigger. The jump row of Table 3 follows this procedure numerically.

4.5.3. Accuracy of the affine rule

Proposition 4 separates the size of the trigger error from the location of the critical weight, which explains why widths stayed close while labels differed.

Proposition 4 (Local trigger approximation and critical-weight

sensitivity). *At the reference project value $V_0 = I$, fix a recorded weight $\alpha \in [0, 1]$.*

- (i) *First-order agreement. The affine and benchmark triggers both expand about the anchor volatility as $V_0^* + (1 - 2\alpha)V^*(\bar{\sigma}_0)h + O(h^2)$. Hence $V_{\text{lin}}^* - V_{\alpha\text{-MEU}}^* = O(h^2)$.*
- (ii) *Quadratic trigger-error expansion. The relative trigger error satisfies $|V_{\text{lin}}^* - V_{\alpha\text{-MEU}}^*|/V_{\alpha\text{-MEU}}^* = C(\alpha)(h/\bar{\sigma}_0)^2 + O((h/\bar{\sigma}_0)^3)$, where $C(\alpha) = K\alpha(1 - \alpha)$ for a positive calibration-dependent constant K . The leading coefficient is largest at $\alpha = \frac{1}{2}$ and vanishes at the endpoint weights.*
- (iii) *Separation of magnitude and location. At a positive radius, the critical weight is defined by $V^*(\tilde{\alpha}, h) = V_0^*$. The ratio of the $O(h^2)$ curvature term to the $O(h)$ slope determines its distance from 1/2. Part (i) puts the trigger discrepancy at order h^2 . Because this discrepancy is divided by the $O(h)$ slope, the critical weights of the two formulations differ at first order, $\tilde{\alpha}_{\text{lin}} - \tilde{\alpha}_{\alpha\text{-MEU}} = O(h)$.*

The local constant in Proposition 4(ii) is $K = \bar{\sigma}_0^2 \Psi_I / V_0^*$, where Ψ_I is the second-order benchmark-curvature term, and K equaled 1.0495 at the baseline calibration.

Over the baseline grid, the benchmark critical weight fell on both sides of one half, depending on the reference project value. It exceeded one half only for reference values below roughly $0.83I$ at a 5% radius and $0.87I$ at 33%. By Corollary 1, the affine critical weight exceeds one half at every positive admissible radius. On the evaluated grid, the two critical weights never coincided at a positive radius, and at $V_0 = I$ they straddled one half. A member with $\alpha_i = 1/2$ was therefore delay-oriented under the affine rule and acceleration-oriented under the benchmark. The weights between the two critical weights form an opposite-label band, which widened with the radius and the reference value. At $V_0 = I$, its width ran from 0.011 at a 5% radius to 0.067 at the 33% stress radius. At $V_0 = 0.5I$, it ran from 0.003 to 0.018.

Whereas part (iii) concerns labels, parts (i) and (ii) of Proposition 4 govern the trigger-level error, which does not pass to the width unchanged. Each member trigger carries an absolute error of $O(h^2)$ and a relative error of $O((h/\bar{\sigma}_0)^2)$. When the extreme weights differ, the width is only $O(h)$, whereas its absolute error

Table 3. Critical weight under three departures from the perpetual geometric Brownian specification.

Specification	Auxiliary grid	Smallest $\tilde{\alpha}$ at $h/\bar{\sigma}_0$			
		5%	10%	22%	33%
Perpetual GBM (reference)	None	0.5086	0.5171	0.5378	0.5570
Finite investment horizon	$T = 5\text{--}80$ years	0.5082	0.5172	0.5376	0.5565
Mean-reverting log price	$\kappa = 0\text{--}0.30$	0.5080	0.5170	0.5375	0.5566
Jump diffusion	$\phi = 0\text{--}0.50$	0.5085	0.5170	0.5375	0.5566

Notes. Each entry is the smallest critical weight over the auxiliary grid at the given radius, so it is the binding case for the above-one-half claim. The finite-horizon grid carries six horizons, the mean-reversion grid four reversion speeds, and the jump grid three variance shares. Across the four radii, the three grids give 24, 16, and 12 cells. In the mean-reversion and jump grids, the settings $\kappa = 0$ and $\phi = 0$, respectively, are solver controls. On the jump grid, the no-jump control bound at every radius, because moving declared variance into jumps raised the critical weight. On the mean-reversion grid, the $\kappa = 0$ control bound at the 10%, 22%, and 33% radii, because a positive reversion speed raised the critical weight there. At the 5% radius, $\kappa = 0.30$ bound by a margin of 0.0005. The perpetual row is the exact affine value of Eq. (6). Each departure is solved by the Crank-Nicolson method on one logarithmic grid shared by the anchor and the two endpoint volatilities. The free boundary is interpolated between nodes.

Table 4. Critical weight and interval width of the three trigger formulations at four relative radii $h/\bar{\sigma}_0$.

$h/\bar{\sigma}_0$	Critical weight $\tilde{\alpha}$			B_t (CNY/t)		
	Affine	Recursive	Pre-commitment	Affine	Recursive	Pre-commitment
5%	0.5086	0.5125	0.4980	5.473	5.494	5.415
10%	0.5171	0.5250	0.4960	10.945	11.027	10.717
22%	0.5378	0.5550	0.4920	24.069	24.442	22.999
33%	0.5570	0.5825	0.4898	36.073	36.861	33.775

Notes. Baseline calibration and four-member reference profile. The recursive critical weight is the closed form $1/2 + \rho/4$ of Eq. (17), which numerical evaluation reproduced to 14 decimal places. The pre-commitment columns are evaluated at $V_0 = I$. Only the pre-commitment output depends on this reference value, and only its critical weight fell below one half.

is $O(h^2)$, or $O(h^3)$ for extreme weights symmetric about one half. Without this symmetry, the relative width error is therefore $O(h/\bar{\sigma}_0)$.

Unlike the width, the directional labels depend on the location of the critical weight, and this dependence leads to two reporting rules. First, a label is released only with the reference project value and the declared ambiguity coordinate. Second, a label is de-emphasized when the recorded weight of the member lies within the illustrative ± 0.05 review band around the critical weight. Elicitation noise alone can move such a weight across the critical weight. In a perturbation test, independent errors with standard deviation 0.05 were added to a ten-member Beta(5, 5) profile. The interval width stayed centered near its noiseless value, with a coefficient of variation of 12.9%. By contrast, the label-flip probability for the weights nearest the critical weight reached 47.6%. At every radius admitted by the release rule, the review band contained the whole opposite-label band. A member whose label depends on the formulation there fore already receives a de-emphasized label. The review band covers only the labels, whereas the contested-price interval and the invest-or-wait classification fall under a separate safeguard on the member thresholds.

The orders in Proposition 4 are local, so Table 5 reports the finite-radius approximation error at the baseline calibration with $V_0 = I$. On the primary radius range, the maximum relative

member-trigger error was 1.26%, or at most 1.47 CNY/t in the marginal carbon-price threshold. On the extended range, the maximum reached 2.82%. Across the 33 admissible calibrations, the maxima rose to 2.74% on the primary range and 6.15% on the extended range. At the largest radius of the reconstructed signal path, the interval-width error for the four-member reference profile was at most 2.25% across tested reference values. Figure 3 maps the relative trigger error over the full grid of radii and weights at the baseline calibration.

Fixing $V_0 = I$ removes the dependence of the benchmark vector on V_0 only through a second reporting choice. The affine rule needs no such choice, since it uses only the two endpoint triggers and the recorded weights. For reference values from $0.75I$ to $1.25I$, the reported affine interval stayed unchanged. The benchmark interval departed from it by at most 0.28 CNY/t inside the release cap. The gate outcome agreed under both rules at every sampled value. The affine report is therefore released, and the benchmark serves as the accuracy standard and the fallback.

4.6. Classification fidelity and information retained by member thresholds

4.6.1. Classification fidelity

Small trigger errors can still change a classification when a price lies between the affine and benchmark thresholds. The comparison is therefore made on the price axis, and two quantities

Table 5. Accuracy of the affine rule against the numerically computed α -MEU benchmark.

$h/\bar{\sigma}_0$	α	V_{lin}^*/I	$V_{\alpha\text{-MEU}}^*/I$	Abs. rel. err. (%)	Signed ΔP^* (CNY/t)
0.10	0.25	3.241	3.235	0.19	+0.23
0.10	0.50	3.144	3.136	0.26	+0.31
0.10	0.75	3.047	3.041	0.20	+0.23
0.22	0.25	3.384	3.355	0.85	+1.07
0.22	0.50	3.170	3.131	1.25	+1.47
0.22	0.75	2.956	2.926	1.02	+1.12
0.33	0.25	3.531	3.470	1.77	+2.30
0.33	0.50	3.211	3.124	2.76	+3.24
0.33	0.55	3.147	3.060	2.81	+3.23
0.33	0.75	2.890	2.824	2.36	+2.50

Notes. Relative errors are unsigned magnitudes. The signed difference is $\Delta P^* = P_{\text{lin}}^* - P_{\alpha\text{-MEU}}^*$, where a positive value indicates a higher marginal carbon-price threshold under the affine rule. The rows show a subset of the α grid. Over the whole grid, the baseline maxima were 1.26% on the primary range and 2.82% on the extended range. Both were attained at weights between the tabulated rows.

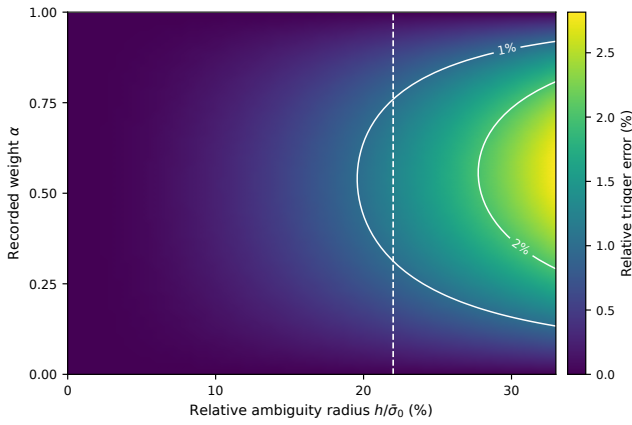


Figure 3. Relative trigger error of the affine rule against the numerically computed α -MEU benchmark. The grid covers the relative ambiguity radius $h/\bar{\sigma}_0$ and the recorded weight α at the baseline calibration and $V_0 = I$. Shading shows the relative trigger error in percent, and white contours mark the 1% and 2% levels. The dashed vertical line marks $h/\bar{\sigma}_0 = 22\%$, the upper bound of the primary radius range. The error peaked near $\alpha = 1/2$ and vanished at the endpoint weights, consistent with Proposition 4(ii).

are computed for each configuration from the affine and benchmark member-threshold vectors. The first is the intersection over union (IoU) of the affine and benchmark contested-price intervals. The second is the omitted mixed share, the length of the benchmark interval outside the affine interval divided by the benchmark width. For prices in this omitted region, the affine rule reports a unanimous classification, whereas the benchmark reports a mixed one. Both diagnostics are geometric, so they require no assumption on the distribution of the prevailing price.

Both diagnostics are evaluated on a fixed deterministic design, with the reference project value equal to the investment cost. The design combines the 33 admissible calibrations with four panel sizes (3, 5, 7, and 9) and six recorded-weight profiles. Crossed with relative radii of 5%, 10%, 15%, and 22%, these factors give 3,168 configurations.

Over this design, Criterion C1 requires at least 95% of configurations to attain an IoU of at least 0.90. Criterion C2 requires the 95th percentile of the omitted mixed share to be at most 5%.

Both criteria therefore concern the tail of the distribution across configurations. On the full set, 89.5% of configurations attained an IoU of at least 0.90, below the 95% required by C1. The 95th percentile of the omitted mixed share was 7.65%, above the C2 limit, so the full set met neither criterion. Fidelity improved at smaller radii, and at the 10% radius alone, 98.2% of configurations met the IoU requirement. On the pooled 5% and 10% set, this share was 99.1%, and the 95th percentile of the omitted mixed share was 4.15%. Both criteria held on this restricted set.

4.6.2. Information retention

Criteria C1 and C2 test whether the affine interval reproduces the benchmark interval. A separate simulation examines whether the affine member-threshold vector retains price-axis information lost by any single-trigger summary of the benchmark vector. The simulation generates 2,000 panels for each combination of five radii, four panel sizes, and five weight families, giving 200,000 panels. Of these, 160,000 lie within the primary radius range. Each simulated panel draws one of the 33 calibrations.

Member-vector loss is the mean absolute price-axis distance between each affine member threshold and its benchmark counterpart, expressed as a percentage of the anchor threshold. Four single-trigger candidates enter the comparison: the anchor threshold, the benchmark thresholds at the mean and median recorded weights, and the largest benchmark member threshold. The loss of a candidate is computed by replacing every affine member threshold with the candidate threshold. For each panel, a four-candidate ex post comparator uses the generated benchmark vector to select the candidate with the lowest loss. This hindsight choice favors the single-trigger summaries in the comparison.

Criterion C3 has two parts. The median member-vector loss must be at most one half of the median loss of the comparator. The four primary radii, four panel sizes, and five weight families define 80 design cells. Each cell contributes the median, over its panels, of the comparator loss minus the member-vector loss. The 95% resampling interval of these cell-level differences must also exclude zero.

Across the panels within the primary radius range, the median member-vector loss was 0.224% of the anchor threshold. The median loss of the comparator was 1.944%, and the resulting 88.5% reduction in median loss met the first part of C3. The member-threshold vector had the lower loss in 99.5% of the panels.

The comparator selected the median-weight benchmark threshold in every tested panel, since the median minimizes an absolute-distance loss and all tested panel sizes are odd. For the second part of Criterion C3, the median cell-level loss difference across the 80 design cells was 1.65% of the anchor threshold. Its 95% resampling interval ran from 1.31% to 2.06% and excluded zero, so Criterion C3 was met.

A separate deterministic grid, equal in size to the fidelity design, examined how much of this gain requires the full vector. A three-threshold summary, reporting the smallest, median, and largest benchmark thresholds, cut the median loss to 0.502% of the anchor threshold. On the same grid, the median loss was 2.240% for the best single-trigger summary and 0.235%

for the member-threshold vector. The full vector thus removed 53% of the loss left by the three-threshold summary and was more accurate in 72.5% of configurations. Three-member panels form a quarter of this grid, and there the three-threshold summary reproduces the benchmark vector exactly. The comparison is therefore informative only on the remaining configurations. Adding a min-max midpoint and a quartile-based composite to the single-trigger candidates left this ordering of losses unchanged. Under both candidate sets, most of the improvement over a single-trigger summary came from reporting more than one threshold. The member-threshold vector also identifies the member behind each threshold, whereas a sorted summary loses this link.

For the Monte Carlo panels, Table 6 reports the results by radius, and each row summarizes 40,000 simulated panels. Its omitted-mixed-share percentiles differed from the deterministic statistics because the simulation drew random weight profiles.

The same panels supply two inputs for an institution setting its own release criteria. The first is a length in price units, the omitted mixed share multiplied by the benchmark interval width. Table 6 reports this length normalized by the anchor threshold, from 0.09% at a 5% radius to 3.60% at the stress radius. At the baseline anchor threshold, these values equaled about 0.10 and 4.24 CNY/t. The second is panel dependence. Fidelity improved monotonically with panel size and degraded under one-sided weight skew. A skewed profile brings the two interval edges closer together, so a given threshold error removes a larger share of the benchmark interval. The weakest cell was a three-member panel under skew. A committee can compare these lengths with the price band over which it regards the gate decision as open. Given the panel dependence, it can then set its own release cap.

This weakest cell is visible at the gate without the benchmark. A dispersion statistic of the recorded weights might therefore serve as an escalation rule for benchmark recomputation. One natural candidate is the ratio of the range of the recorded weights to their interquartile range. This ratio failed as an escalation rule. For three-member panels, it equals two for every profile. Their classification fidelity nonetheless ranged from 48.5% to 99.2%, measured as the share of configurations with an IoU of at least 0.90. For larger panels, the correlation between the ratio and fidelity stayed below 0.25. The weight profile and the panel size did separate passing cells from failing cells, and a gate records both. The release rule nonetheless conditions on the distances between the prevailing price and the thresholds, because these distances catch the affected gates directly.

Beyond the simulated panels, the four-member reference profile was evaluated deterministically in 30 direct comparisons with the benchmark, over five radii and six reference values. These checks and the 32 archived-price cells of Table 13 are included in the replication package.

4.6.3. Finite-registry safeguard

Within the tested design, the simulation supported retaining the member-threshold vector. Classification fidelity near the interval edges, however, weakened as the radius grew, most sharply for skewed profiles and small panels. A release rule is therefore attached to the member-threshold report. Consider

a registered calibration $\theta = (r, \mu_c, \bar{\sigma}_0)$ and an evaluated radius $\rho = h/\bar{\sigma}_0$. With L_{lin} and U_{lin} as the edges of the affine interval C_t , let

$$\begin{aligned} e_{\mathcal{R}}(\theta, \rho) &:= \max_{\alpha \in [0,1]} \frac{|V_{\text{lin}}^*(\alpha) - V_{\alpha\text{-MEU}}^*(\alpha, I)|}{V_0^*}, \\ E_{\mathcal{R}}(\theta, \rho) &:= P_0^* [e_{\mathcal{R}}(\theta, \rho) + \varepsilon_{\text{num}}], \\ d_{\text{edge}} &:= \min\{|P_t - L_{\text{lin}}|, |P_t - U_{\text{lin}}|\}, \\ d_{\text{member}} &:= \min_i |P_t - P_{i,\text{lin}}^*|. \end{aligned} \tag{18}$$

The distance d_{edge} protects the three-state gate summary of unanimous wait, mixed, or unanimous invest, whereas d_{member} protects the invest-or-wait classification of every member. The interval edges are member thresholds, so $d_{\text{member}} \leq d_{\text{edge}}$ and the member condition binds, although both distances are retained for audit. Each distance is compared with the envelope $E_{\mathcal{R}}$, which converts the registered maximum anchor-normalized member-trigger error plus a numerical tolerance into price units. Because P_0^* multiplies both $e_{\mathcal{R}}$ and the tolerance, a common rescaling of the member thresholds rescales $E_{\mathcal{R}}$ by the same factor. The anchor-normalized trigger error is invariant, yet the release decision can still change, because the prevailing price stays fixed while the thresholds rescale. The registry of evaluated configurations therefore stores the dimensionless envelope $e_{\mathcal{R}}$. After any revision of the investment cost or the captured volume, the gate recomputes P_0^* and $E_{\mathcal{R}}$ and re-evaluates both proximity distances. The audit record carries I/Q and P_0^* with the registry identifiers, so a past release can be reconstructed on its original price axis.

The registry decides only when the benchmark may be omitted, and a committee may still request the benchmark at any gate. Affine-only reporting requires four conditions. (i) The calibration and the canonical radius key are registered exactly. (ii) The radius is at most 10%. (iii) The reference value satisfies $V_0/I = 1$, the only value at which the release rule was evaluated. (iv) Both proximity distances exceed $E_{\mathcal{R}}$. If any condition fails, or if the solver or envelope version is unregistered, the gate recomputes the benchmark and issues a dual report. In either mode, the reporting interface passes on the member-threshold vector and the fallback status. A registry match under condition (i) must come from the frozen serialized key. Rounded display values and values reparsed from the upstream signal file do not qualify. Interpolation is excluded, because no certified release envelope exists between evaluated radius keys. The 10% radius cap is a hard cutoff, so rounding a configuration constant can move a radius to the wrong side of it. One low-radius scenario targeted exactly 10% and yielded 10.30% at its rounded constant, outside the affine-only region. Exact-key matching guards against this error. An institution may prespecify a rule mapping each gate radius to a registered key, and the release guarantee then applies to the discretized ambiguity set. The audit record retains both the original radius and the registered radius.

When the benchmark is recomputed, the gate first evaluates the uniqueness condition (14). When this sufficient condition holds, the maximizer is the unique stationary point on $[V_{\text{low}}^*, V_{\text{high}}^*]$, and a bounded solve suffices. When it fails, the gate isolates every stationary point and compares the objective at

Table 6. Classification fidelity and price-axis information loss across simulated panels of sizes 3, 5, 7, and 9. The 33% radius is a stress case.

$h/\bar{\sigma}_0$	IoU, median (5th percentile)	95th-percentile omitted mixed share / anchor-normalized omitted length (%)	Median loss, member vector / four-candidate ex post comparator (% of anchor threshold)
5%	0.978 (0.929)	4.0 / 0.09	0.039 / 0.862
10%	0.957 (0.862)	7.9 / 0.35	0.156 / 1.720
15%	0.937 (0.801)	11.7 / 0.78	0.350 / 2.601
22%	0.911 (0.723)	16.8 / 1.65	0.735 / 3.810
33%	0.876 (0.621)	23.9 / 3.60	1.573 / 5.746

each and at both endpoints before releasing a benchmark classification. Condition (15) is recorded separately, because it governs the one-sided reading of the dual report. Algorithm 1 states the matching, versioning, and solver rules.

The registry makes a released classification auditable after the fact. Each registry cell is indexed by the economic calibration, the radius key, and the reference value, and it carries the envelope and solver versions. The gate audit record also holds the recorded weights and the prevailing price. A later reviewer can then identify the inputs behind a past classification and recompute it without re-eliciting any weight. In prospective use, the registry should be append-only, so a revised input opens a new dated cell. The earlier cell then stays readable with its original guarantee. The release rule also isolated the configurations in which the two formulations disagreed. Across the 3,960 registered configurations, such disagreement occurred in 57. None of these 57 passed the radius cap and both proximity conditions together. Two of them lay at a 10% radius, where a proximity flag routed them to benchmark recomputation. The envelopes behind these flags were constructed at all 3,960 domain-valid configurations with $V_0/I = 1$, and two checks bounded their numerical error. First, a denser α grid detected no exceedance of the registered envelope in any of the 165 calibration-radius cells. Second, the maximum on the finite weight grid was compared with a certified upper bound on the discrepancy over $[0, 1]$ in each cell. This bound used a closed-form Lipschitz constant and a branch-and-bound search. It exceeded the grid maximum by at most 3.1×10^{-6} CNY/t and changed no release decision.

The 10% radius cap in condition (ii) is the largest evaluated radius at which both fidelity criteria held in the deterministic design. A refinement at intermediate radii located the failure just above the cap. C2 failed first, at 11%, where the 95th percentile of the omitted mixed share reached 5.4% against the 5% limit. At the 10% radius, the same quantity was 4.96%, so C2 held at the cap by 0.04 percentage points. The breach at 11% was 0.4 percentage points, which suggests a gradual loss of accuracy across the cap. By Proposition 4(i), the leading error term grows with the squared radius, so raising the cap to 22% would multiply it by about five.

The cap was also tested on 18 calibrations and five weight profiles absent from the evaluated grid, six of these calibrations lying outside its range. The price-axis quantity used for release behaved similarly on this held-out set. The 95th percentile across cells of the largest within-cell member-threshold deviation was 0.5141 CNY/t, against 0.5143 CNY/t on the evaluated grid. The normalized fidelity criteria were more sensitive to the weight profile and held on four of the five held-out profiles. They failed

on one profile with weights clustered within 0.06 of one half, for which the benchmark interval shrank to 0.72% of the anchor threshold. A similar absolute discrepancy then became a much larger relative one. Both criteria registered this effect, because IoU normalizes by the union of the two intervals and the omitted mixed share by the benchmark width. For this reason, the release rule conditions on absolute price-axis distances.

Above the cap, the dual report is mandatory. Across the three radii above the cap and the weakest profiles, member-level classifications agreed in more than nine cases in ten. The remaining differences ran in one direction. In no evaluated cell did the affine rule classify a price as invest while the benchmark classified it as wait. Proposition 3(ii) explains this pattern, because under the one-sidedness condition the affine report errs toward deferral. The condition held throughout the affine-only region as well, so any residual affine error there also leans toward deferral.

Figure 4 shows the release decision as a single pass over the four release conditions. When all four hold, E_R bounds the error of every member threshold, so the released classification matches the benchmark classification. The released interval width B_r , however, is bounded only in absolute terms, within $2E_R$. In relative terms, the width error for the four-member reference profile was 1.29% at the reference project value of the release rule. The 2.25% maximum across tested reference values arose outside the release region. The width shrinks as the recorded weights cluster, so the same absolute error is a larger fraction of the width on tightly clustered profiles. In Figure 4, input validity is checked on a separate branch, and a failed upstream acceptance check blocks the threshold report.

With the release rule in place, the model results fall into two groups with different scopes. The analytical group, covering the critical weight and its location relative to one half, holds wherever the maintained model holds. The numerical group, covering the accuracy of the affine rule and its release conditions, holds only on the finite registry of evaluated configurations. The practical relevance of both groups depends on the inputs of a specific project.

5. Illustrative numerical application to a 600-MW CCUS retrofit

A 600-MW coal-fired CCUS retrofit supplies the inputs of a specific project for this illustration. The application combines the economic calibration of this retrofit with an ambiguity radius from a date-reconstructed policy-text signal. A four-member reference profile of recorded weights and the archived price

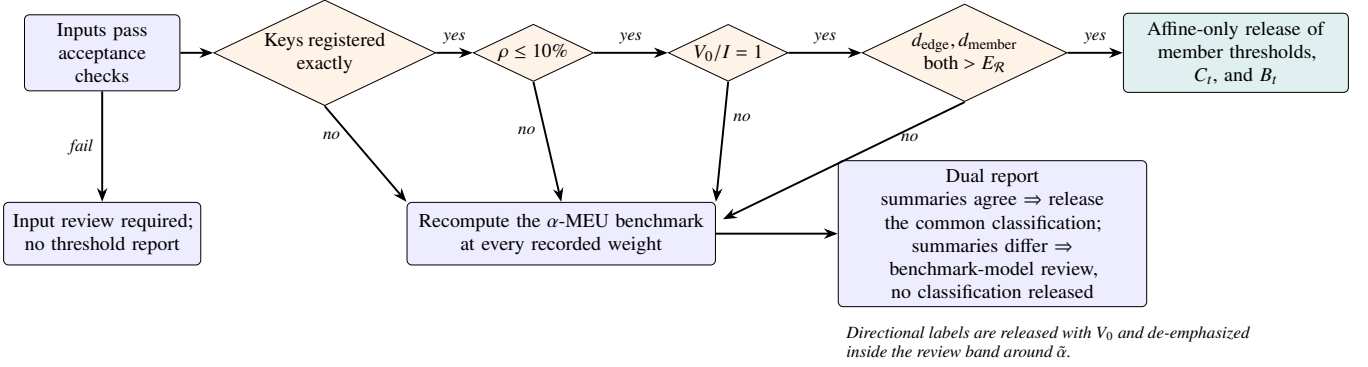


Figure 4. Release decision for one gate. The gate first runs the registered input acceptance checks, and a failure blocks the threshold report. Four release conditions are then checked in sequence. The calibration and radius key must be registered exactly, and the relative radius must be at most 10%. The reference project value must satisfy $V_0/I = 1$, and both d_{edge} and d_{member} must exceed the envelope E_R . Passing all four conditions releases the affine-only report with the member-threshold vector, C_t , and B_t . Failing any condition routes the gate to a dual report, with the α -MEU benchmark recomputed at every recorded weight. In this mode, both member-threshold vectors are issued. A gate-level classification is released only when the two three-state summaries agree, although individual member classifications may still differ. A discrepancy between the summaries leads to a benchmark-model review, with no precedence for either rule.

series complete its four inputs. From these inputs, the method re-
turns member thresholds and quarterly contested-price intervals
against which each archived daily China Emission Allowance
(CEA) price is classified. The classification is retrospective
because the anchor, corpus, and encoder are fixed ex post. For
example, the volatility anchor estimated on 2024 and 2025 data
applies to the entire historical signal path. Sensitivity checks
varied the weight profiles, economic inputs, volatility anchors
and scaling constants.

5.1. Numerical setting and policy-text signal

5.1.1. Corpus and signal construction

The corpus comprises 682 central-government energy and
climate policy records from four issuing bodies. These bodies
are the National Development and Reform Commission, the
National Energy Administration, the State Council, and the Min-
istry of Ecology and Environment. Of these records, 678 fell in-
side the analysis window from 2016Q1 to 2025Q4. The records
were retrieved from the PKULaw database, which assigns each
one a stable document number.

The retrieval targeted the years 2016 to 2025. A thematic
screen based on source tags and a title keyword filter retained
records classified under energy, climate, carbon markets, or
emissions trading. A document-type screen kept substantive
plans, notices, guidance, and formal responses while excluding
press releases, routine administrative notices without policy
content, and non-Chinese texts.

The policy-text signal is quarterly, so an incorrect document
date can place a record in the wrong quarter and distort the signal
path. Dates in the database may differ from official issuance
dates. An audit therefore matched each record to an official
agency page where possible and located such a page for 414
of the 682 records. Of the two date assignments compared
the reference assignment uses the date printed in each archived
record and is reproducible from the archive alone. The page
date assignment uses a verified agency page date where available
and the printed date otherwise. The two assignments produced
different paths for the quarterly signal and the interval width

Window membership also follows the printed dates: four records
bear 2015 dates and fall outside the window, so they are excluded.
Four other records carry month-only dates inside the window
and are retained, because a month identifies the quarter.

Each document is represented by precomputed layer-14
embeddings from the Qwen3-0.6B model (Yang et al., 2025).
The embeddings are pooled across subword tokens with term
frequency-inverse document frequency (TF-IDF) weights
and normalized to unit length. This primary representation,
Qwen3-L14, was fixed before the signal was compared with
realized volatility. From these unit vectors, the policy-semantic-
dispersion index is

$$\text{PSDI}_t = 1 - \|\bar{c}_t\|, \quad (19)$$

where $\bar{c}_t$ is the centroid of all normalized document vectors in
period t . A larger PSDI_t indicates greater dispersion around
the period centroid. The rolling variability signal tracks the
temporal variation of this dispersion through the four-quarter
standard deviation

$$w_t = \text{std}_4(\text{PSDI}_t, \text{PSDI}_{t-1}, \text{PSDI}_{t-2}, \text{PSDI}_{t-3}). \quad (20)$$

Here std_4 is the population standard deviation, with divisor four,
over the current and previous three quarters. Figure 1 abbreviates
this window as $\text{PSDI}_{t:t-3}$.

Replacing the dispersion index by six alternative statistics on
the same vectors moved the peak quarter in two cases, although
none changed the gate outcome. One corpus-composition check
moved both the peak quarter and the peak width. This check
excluded documents longer than 2,000 characters before trun-
cation and retained 384 of the 678 records. The peak quarter
shifted from 2024Q4 to 2023Q1, and the peak interval width
rose from 6.63 to 8.59 CNY/t. The Spearman rank correlation
between this length-filtered path and the reference path was 0.61.
One of two replacement encoders, MacBERT-base, also moved
the peak, placing it in 2019Q2 at a width of 5.42 CNY/t. Its
quarterly signal had a Spearman rank correlation of 0.305 with
the reference path.

5.1.2. Screening procedure and input diagnostics

Algorithm 1 summarizes the gate update from the registered radius source through the release rule to the reported object. Its computational cost depends on the radius source, and with a declared radius the affine branch is $O(m)$ in the panel size m . With the text-derived radius, it is $O(|D_t| + m)$ once the embeddings exist, where $|D_t|$ is the number of documents in period t . This bound assumes a fixed embedding dimension and a four-quarter history. The fallback adds one benchmark optimization per member. Computing the release envelope of one calibration-radius cell on the 2,001-point weight grid took about 0.2 seconds, a wall-clock figure varying with machine load.

In the text branch of Algorithm 1, the proportional map $h_t = c_w w_t$ requires a nonnegative signal, since h_t is a radius. The rolling standard deviation is nonnegative by construction. The reference configuration sets the scaling constant c_w to one, so the ambiguity radius equals the signal before the admissibility cap. An estimated c_w would need a stable association between the lagged signal and realized volatility. Before the national emissions trading scheme (ETS), this association was positive but imprecise. Over the 18 quarters of the national ETS period the estimated slope turned negative at -18.07 , with a 95% confidence interval of $[-34.57, -1.57]$. The confidence interval for the shorter estimation window spanned zero. Neither subsample identified a stable value for c_w . Five further diagnostics, covering directional association, temporal-order permutation, construct validity, an event-style check, and national ETS stability, supported no stable mapping either. The signal therefore serves as a scenario input configuring the radius.

Slope estimates distinguishable from zero, such as the national ETS slope, still left the scale of the radius open. Propagating the pre-ETS slope to the thresholds gave a width envelope of up to 19.2 CNY/t. This value was about 2.9 times the configured peak width of 6.63 CNY/t. The pre-ETS slope and the national ETS slope differed by 61.08, with a standard error of 30.94. The 95% confidence interval for this difference, based on the t distribution with 23 Welch degrees of freedom, contained zero. With slopes of opposite sign and a difference indistinguishable from zero, the data did not fix even the direction of a rescaling. A mapping re-estimated on the national ETS window explained none of the out-of-sample variation in 2024 and 2025. Separately the signal was compared with an established policy-uncertainty measure, the log China Economic Policy Uncertainty (EPU) index of Baker et al. (2016). Over 2017 to 2024, the Pearson correlation was 0.265 ($n = 32$), with a four-lag heteroskedasticity and autocorrelation consistent (HAC) p -value of 0.036. Over the 14 national ETS quarters covered by the EPU sample, it was 0.264.

Although the slope estimates left the radius scale open, two properties suit the signal to its illustrative role. First, an institution can archive and re-encode the documents. With the corpus encoding pipeline, and scaling convention recorded, a past radius can be recomputed without re-eliciting any expert input. Second, the documents come from the same regulatory process whose ambiguity the committee evaluates at each gate.

Two calibration routes could replace the scaling conven-

tion with an estimate. A forward-looking route would set the endpoints from option-implied volatility bounds. This route requires a traded allowance option, and the national market had none over the sample period. A survey route would use the forecast dispersion elicited from allowance-price forecasters with a documented track record. Each source would need its own validation and an explicit mapping to the declared radius. Combining the two would need an aggregation rule for their overlapping information.

For prospective use, an institution should set a calibration cutoff and use only information available by this date. Before later gate outcomes are observed, it should also register a target radius h_{ref} and a strictly positive reference signal w_{ref} , which fix $c_w = h_{\text{ref}}/w_{\text{ref}}$. The registration also covers the volatility-anchor window, the encoder version, the admissibility cap, and the review cadence. Later gate outcomes do not retune c_w or the reference statistic. A documented drift or regime review opens a new dated configuration under the append-only rule. The target radius, achieved radius, rounded display value, and canonical registry key are distinct fields, because release matching follows the exact-key rule. Under this rule, an unseen radius sends the gate to benchmark recomputation. Prospective affine-only use therefore requires a radius discretization declared in advance or a release envelope certified over a continuous radius interval. A stable forward-volatility series may later support statistical calibration. The institution should then prespecify training and holdout windows and register the fitted map as a new configuration.

5.2. Baseline setting

Table 7 lists the baseline calibration for the 600-MW retrofit. The investment cost and annual captured emissions follow Sun et al. (2024). The discount rate follows the Chinese national appraisal standard (National Development and Reform Commission [NDRC] and Ministry of Construction, 2006). The annual carbon-price growth scenario is set to 0.04. The volatility anchor is the annualized standard deviation of 474 within-market log returns from 475 daily national CEA closing prices in 2024 and 2025. The Shanghai pilot and national CEA data are separate series, so no return in any anchor window spans the market boundary. No liquid CEA options market existed over the sample period, so the historical estimate of 0.249 serves as the forward-volatility anchor.

This calibration gave an anchor threshold of 117.7 CNY/t, and Table 8 places it beside external capture-cost references, including a demonstration reference near 250 CNY/t. P^* is a marginal threshold, whereas the references are technology costs. A committee comparing the two therefore first shifts the reported basis by a common per-tonne adjustment. Table 9 evaluates the common-adjustment identity at illustrative values of Δ spanning these references. In a hypothetical accounting scenario with $\Delta = 250$ CNY/t, the interval moved to [363.38, 370.01) CNY/t.

The per-tonne cost adjustment enters the thresholds additively, whereas the calibrated inputs I and Q enter multiplicatively. By Eqs. (2) and (8), every P_i^* is proportional to I and inversely proportional to Q . A different investment-cost or

Algorithm 1 Gate-level member thresholds and safeguarded classification.

Input One registered radius source, either a declared radius h_t (or ρ_t) or the application-specific text construction with its registered pipeline identifiers

Input Dated deployment record for $\bar{\sigma}_0$, c_w , ϵ , the registered price-recording buffer b , and registry-versioning rules

Input Recorded weights $\{\alpha_i\}_{i=1}^m \in [0, 1]^m$ passing the prespecified input-acceptance checks ▷ reporting inputs

Input $r > \mu_c > 0$, $I, Q > 0$, prevailing price P_t , and reference project value V_0 inside the benchmark domain

Input Versioned finite registry snapshot $\mathcal{R}^{(v)}$ with frozen canonical serialized configuration identifiers

Input Direct-benchmark request flag $b_{\text{direct}} \in \{0, 1\}$

Output Affine member table, $\tilde{\alpha}$, C_t , B_t , released classification, audit record, and reporting mode

Output Benchmark table on fallback or direct request

1. initialize $e_{\mathcal{R}}$, $E_{\mathcal{R}}$, both proximity flags, and both certification flags to *not evaluated* ▷ filled when the lookup or check runs
2. **if** the deployment record, economic, or recorded-weight checks fail **then**
3. **return** input review required; issue no threshold report
4. **end if**
5. **if** the registered radius source is a declared radius **then**
6. read the registered h_t or ρ_t
7. **else**
8. obtain h_t by the application-specific text construction; Eqs. (19) and (20) give PSDI _{t} and w_t , and the admissibility cap gives h_t ▷ application-specific step
9. **end if**
10. **Analytical layer.** $\rho_t \leftarrow h_t / \bar{\sigma}_0$; $(\sigma_-, \sigma_0, \sigma_+) \leftarrow (\bar{\sigma}_0 - h_t, \bar{\sigma}_0, \bar{\sigma}_0 + h_t)$
11. solve Eq. (3) at $(\sigma_-, \sigma_0, \sigma_+)$ and compute $(V_{\text{low}}^*, V_0^*, V_{\text{high}}^*)$
12. **if** $V_0 \geq V_{\text{low}}^*$ **then**
13. **return** input review required; issue no threshold report ▷ benchmark-domain check
14. **end if**
15. $V_i^* \leftarrow \alpha_i V_{\text{low}}^* + (1 - \alpha_i) V_{\text{high}}^*$ for all i
16. $\tilde{\alpha} \leftarrow 1/2$ if $h_t = 0$; otherwise $\tilde{\alpha} \leftarrow (V_{\text{high}}^* - V_0^*) / (V_{\text{high}}^* - V_{\text{low}}^*)$
17. **Interface layer.** $P_0^* \leftarrow V_0^* (r - \mu_c) / Q$; $P_i^* \leftarrow V_i^* (r - \mu_c) / Q$ for all i ; $C_t \leftarrow [\min_i P_i^*, \max_i P_i^*]$; $B_t \leftarrow \max_i P_i^* - \min_i P_i^*$
18. affine summary $\leftarrow \text{CLASSIFY}(P_t, \{P_i^*\})$ ▷ three-state gate summary
19. $d_{\text{edge}} \leftarrow \min\{|P_t - \min_i P_i^*|, |P_t - \max_i P_i^*|\}$; $d_{\text{member}} \leftarrow \min_i |P_t - P_i^*|$
20. buffer-flag $\leftarrow [\min\{d_{\text{edge}}, d_{\text{member}}\} < b]$ ▷ price-input diagnostic reported beside the classification
21. $(k_\theta, k_\rho, k_V) \leftarrow \text{CANONICAL_IDS}(r, \mu_c, \bar{\sigma}_0, \rho_t, V_0 / I)$
22. registered $\leftarrow \text{REGISTRY_LOOKUP}(k_\theta, k_\rho, k_V; \mathcal{R}^{(v)})$ ▷ exact values; no rounded match or interpolation
23. **if** registered **then**
24. read the dimensionless registered envelope $e_{\mathcal{R}}$, the grid maximum recorded for the cell ▷ I and Q enter only through P_0^*
25. $E_{\mathcal{R}} \leftarrow P_0^* [e_{\mathcal{R}} + \epsilon_{\text{num}}]$; near-edge $\leftarrow [d_{\text{edge}} \leq E_{\mathcal{R}}]$; near-member $\leftarrow [d_{\text{member}} \leq E_{\mathcal{R}}]$
26. fallback $\leftarrow b_{\text{direct}} \vee [V_0 / I \neq 1] \vee [\rho_t > 0.10] \vee \text{near-edge} \vee \text{near-member}$
27. **else**
28. both proximity flags $\leftarrow$ not evaluated; fallback $\leftarrow$ true
29. **end if**
30. **if** fallback **then**
31. global-certified $\leftarrow$ Condition (14) holds at (θ, ρ_t)
32. **if** global-certified **then**
33. solve Eq. (13) on $[V_{\text{low}}^*, V_{\text{high}}^*]$ at (α_i, V_0) for all i
34. **else**
35. isolate every stationary point of Eq. (13) on $[V_{\text{low}}^*, V_{\text{high}}^*]$ and compare the objective at all of them and at both endpoints
36. **if** no global maximizer is certified **then**
37. **return** benchmark-model review required; issue no released classification
38. **end if**
39. **end if**
40. one-sided-certified $\leftarrow$ Condition (15) holds at (θ, ρ_t) ▷ governs the one-sided reading only
41. convert the benchmark triggers by Eq. (8)
42. mode $\leftarrow$ dual report
43. **if** the affine and benchmark three-state summaries agree **then**
44. released classification $\leftarrow$ this summary ▷ agreement concerns the summary; member assignments may differ, and both member tables are released
45. **else**
46. released classification $\leftarrow$ none; benchmark-model review required ▷ no priority for either rule
47. **end if**
48. **else**
49. released classification $\leftarrow$ affine summary; mode $\leftarrow$ affine-only report
50. **end if**
51. record the audit fields I / Q , P_0^* , $e_{\mathcal{R}}$, and $E_{\mathcal{R}}$, the radius source, the two certification flags, and the mode
52. **return** affine member table, C_t , B_t , $\tilde{\alpha}$, released classification, $(v, k_\theta, k_\rho, k_V)$, the audit record, and the benchmark table on fallback or direct request

1346 captured-volume assumption is therefore a common-factor ad₃₅₁
1347 justment. The trigger-to-cost ratios stay unchanged, because₃₅₂
1348 V^* is proportional to I and independent of Q . The threshold₃₅₃
1349 levels, the interval width, and the price-unit envelope rescale₃₅₄
1350 with the common factor, so the classification at a fixed prevailing₃₅₅

price may move. Both inputs may carry considerable appraisal
uncertainty. The 2024Q4 peak of the reconstructed reference
path, at a radius of 6.057% of the volatility anchor, shows the
effect of this uncertainty. At this radius, the baseline lower inter-
val edge of 113.38 CNY/t lay above the highest archived price

of 105.65 CNY/t. A 10% reduction in I/Q moved the edge to 102.04 CNY/t, so the same archived price entered the interval and the gate became mixed. A 20% reduction moved the edge to 90.70 CNY/t and placed the price above the whole interval. The critical weight and the member ordering stayed unchanged in all these cases.

5.3. Screening outputs under alternative component configurations

Table 10 compares four configurations, which switch the text-derived radius and the heterogeneous recorded weights on or off under a common economic calibration. The first two configurations yield a single trigger, because the first sets $h = 0$ and the second gives all members the same weight. The third fixes the relative radius at 10% of the volatility anchor, so its member-level interval has the same width at every gate. The full method updates the interval at each gate, and its quarterly widths ranged from 1.67 to 6.63 CNY/t under the reference scaling convention.

5.4. Member-level interpretability at a review gate

The reconstructed path stayed below the 10% release cap, so its intervals were narrower than at the cap. Across the 37 quarterly gates, the relative radius ran from 1.524% to 6.057%. For release, the binding constraint was exact-key matching. The 2024Q4 peak radius was a registered key, and the other 36 gate radii were not. The release rule therefore sent these 36 gates to benchmark recomputation. With a radius discretization registered in advance, an institution could round all 37 radii to registered keys at or under the cap.

At the cap, the upper limit of affine-only eligibility, Table 11 reports the member triggers and thresholds. The recorded weights of the reference profile, from 0.25 to 1.00, are design inputs for the illustration. Its weight range of 0.750 was the widest among five profiles compared at a common radius. Under the affine, strictly decreasing map, it therefore produced the largest interval width among the five. Within this profile, the weight $\alpha = 0.50$ lay just under the critical weight. The affine rule labeled this weight delay-oriented by a narrow margin, and the benchmark labeled it acceleration-oriented. The thresholds, by contrast, changed little across formulations. At the reference peak radius, the affine, recursive, and pre-commitment widths were 6.630, 6.660, and 6.545 CNY/t. The recursive and pre-commitment widths differed from the affine width by +0.46% and -1.27%. A committee may therefore read the thresholds first and treat the directional label as secondary.

5.5. Dynamic intervals and archived-price classification under the reference configuration

Whereas Table 11 fixes the radius, Figure 5 lets it vary across review gates and shows the contested-price interval under the reference configuration. The volatility anchor stays fixed at its estimated value, so all width variation comes from the reconstructed radius path. The shaded band spans the two extreme member thresholds, and its width reached 6.63 CNY/t in 2024Q4. The lower panel compares the observed daily CEA price with

the lower interval edge on the same time axis. To check the retrospective anchor, an as-of anchor estimated only from prices available at each gate was computed at the 14 feasible gates. Over these gates, the peak quarter and the unanimous-wait outcome stayed unchanged.

5.5.1. Archived-price classification

Under the reference configuration, $\Delta = 0$, and the prevailing price equals the archived CEA price. Across the 1,073 daily CEA observations from 2021Q3 to 2025Q4, both the affine rule and the α -MEU benchmark returned unanimous wait. The two rules share the lower interval edge by construction, since the reference profile includes the endpoint weight $\alpha = 1$. The highest archived price, 105.65 CNY/t, lay 7.73 CNY/t below the lowest member threshold along the reconstructed path. The reference configuration is therefore a nonbinding baseline: retaining the member-threshold vector and its interval left the three-state classification unchanged along this price path. By contrast, changing the radius alone, or one economic primitive alone, produced binding configurations. Of 32 sensitivity configurations evaluated against the same price record, 15 placed the highest archived price at or above the lower edge. Ten of these were released as mixed and four as unanimous invest. The remaining one went to benchmark-model review because the two rules disagreed.

5.6. Calibration sensitivity and safeguard behavior

The sensitivity configurations perturb the baseline calibration against the same archived price history, so any change in classification comes from the threshold side.

5.6.1. Worked gate reports

Table 12 renders two cells of the 32-cell sensitivity design as complete gate reports, which exercise the two branches of the release rule. Both use the lower volatility anchor $\bar{\sigma}_0 = 0.199$, the four-member reference profile, and $V_0 = I$, and they differ only in the evaluated radius.

In Configuration A, the radius was the frozen reference peak, and the interval was 4.56 CNY/t wide. The prevailing price exceeded its upper edge by 0.002 CNY/t, so all four members classified this price as invest and the gate was unanimous. The member with the smallest recorded weight holds the upper edge, so the edge distance equals the nearest-member distance. This distance of 0.002 CNY/t was far below the registered envelope of 0.059 CNY/t. The proximity conditions therefore failed, and the gate issued a dual report after benchmark recomputation. The benchmark placed the upper threshold at 105.60 CNY/t, against an affine value of 105.65 CNY/t, and the price cleared both. The proximity rule thus acted conservatively, since the recomputation confirmed unanimous investment. In Configuration B, at the 10% radius, the interval widened to 7.53 CNY/t and contained the unchanged prevailing price. Three members classified the price as invest, whereas the member with the smallest recorded weight classified it as wait, so the gate was mixed. Both proximity distances rose to 1.092 CNY/t, above the envelope of 0.160 CNY/t, so the affine-only report was released.

Table 7. Baseline calibration for the 600-MW CCUS numerical application.

Parameter	Symbol	Value	Source or basis
Investment cost	I	1.585×10^9 CNY	Sun et al. (2024)
Discount rate	r	0.08	NDRC and Ministry of Construction (2006)
Annual carbon-price growth scenario	μ_c	0.04	Decision-analysis input, varied in the sensitivity analysis
Carbon-price volatility anchor	$\bar{\sigma}_0$	0.249	National CEA daily closing prices traded on the Shanghai Environment and Energy Exchange, 2024 to 2025 (475 prices)
Annual captured emissions	Q	1.69×10^6 t/yr	Sun et al. (2024)
Admissibility margin	ϵ	0.5	Fixed cap parameter, nonbinding in the reference application
Signal-to-radius scaling constant	c_w	1.0	Fixed reference convention

Table 8. External cost context for the marginal carbon-price threshold P^* .

Reference quantity	Reported value	Interpretation
This paper, no-ambiguity marginal carbon-price threshold	≈ 117.7 CNY/t	Marginal carbon-price threshold implied by the anchor trigger. It excludes capture operating costs, the energy penalty, transport, storage, and subsidies.
China coal-power CCUS demonstration reference	≈ 250 CNY/t	Reported capture-cost reference for the Taizhou coal-fired carbon-capture facility of China Energy (Yang, 2023). Technology-cost figure.
IEA dilute-stream capture-cost reference	More than USD 40/t, and more than USD 100/t in some cases	Capture-cost range reported by International Energy Agency [IEA] (2020) for dilute CO ₂ streams in power generation and blast-furnace steelmaking.

Table 9. Common per-tonne cost translation of the member thresholds at the reference-peak radius.

Scenario	Δ (CNY/t)	Lower edge (CNY/t)	Upper edge (CNY/t)	B_t (CNY/t)	Margin to $P_{\max}$ (CNY/t)
No common adjustment (reported basis)	0	113.38	120.01	6.63	+7.73
Illustrative low common cost	150	263.38	270.01	6.63	+157.73
Illustrative cost at the Taizhou reference level	250	363.38	370.01	6.63	+257.73
Illustrative high common cost	350	463.38	470.01	6.63	+357.73
Illustrative cost at the Taizhou level, net of an 80 CNY/t subsidy	170	283.38	290.01	6.63	+177.73

Notes. The first row is the reported basis used for every other result in this paper. The remaining rows apply the common-adjustment identity at illustrative Δ values spanning the external references of Table 8. The identity holds for adjustments common across members and constant per tonne. $P_{\max}$ is the highest archived national CEA closing price, 105.65 CNY/t. The final column is the lower edge minus $P_{\max}$, and a positive value means every member classifies this price as wait.

Table 10. Screening outputs under alternative component configurations.

Configuration	Text-derived radius	Heterogeneous recorded weights	B_t (CNY/t)	Output at each gate
Conventional real-options baseline ($h = 0$)	No	No	0	Single trigger
Homogeneous $\alpha = 0.5$	Yes	No	0	Single trigger
Fixed h ($h/\bar{\sigma}_0 = 10\%$, static)	No	Yes	10.945	Fixed interval; classification evaluated at each gate
Full method ($c_w = 1.0$, time-varying)	Yes	Yes	1.67 to 6.63	Time-varying interval and classification summary

Table 11. Member-specific trigger values and ambiguity premiums at the baseline economic calibration with $h_t/\bar{\sigma}_0 = 10\%$.

Case	V^*/I	P^* (CNY/t)	AP/I	Shift relative to anchor
No-ambiguity anchor	3.138	117.7	0	Baseline
$\alpha = 0.25$	3.241	121.6	+0.104	Delay-oriented
$\alpha = 0.50$	3.144	118.0	+0.007	Delay-oriented
$\alpha = 0.75$	3.047	114.3	-0.091	Acceleration-oriented
$\alpha = 1.00$	2.950	110.7	-0.188	Acceleration-oriented

Notes. AP/I is the ambiguity premium divided by the investment cost. The final column labels the shift of each trigger relative to the anchor trigger. The ambiguity premium changed sign at the critical weight, 0.517 at this radius and calibration. The reconstructed radius stayed below 10%, and the lower interval edge decreases with the radius. The value 110.7 CNY/t is therefore a lower bound for the lower edge along the reconstructed reference path.

In this pair, the wider interval permitted affine-only release, because the release test depends on the distance from the prevailing price to the nearest threshold. Widening carried the upper threshold past the price and raised this distance from 0.002 to 1.092 CNY/t.

5.6.2. Margins across the evaluated configurations

Beyond the two worked cells, Table 13 evaluates the baseline calibration, five economic-parameter perturbations, and two additional volatility anchors from Table 14. For each cell, the table reports the signed margin of each interval edge, defined as the edge minus the highest of the 1,073 archived observations. This price minimizes both signed margins, whereas the release test uses absolute distances, which the highest price need not

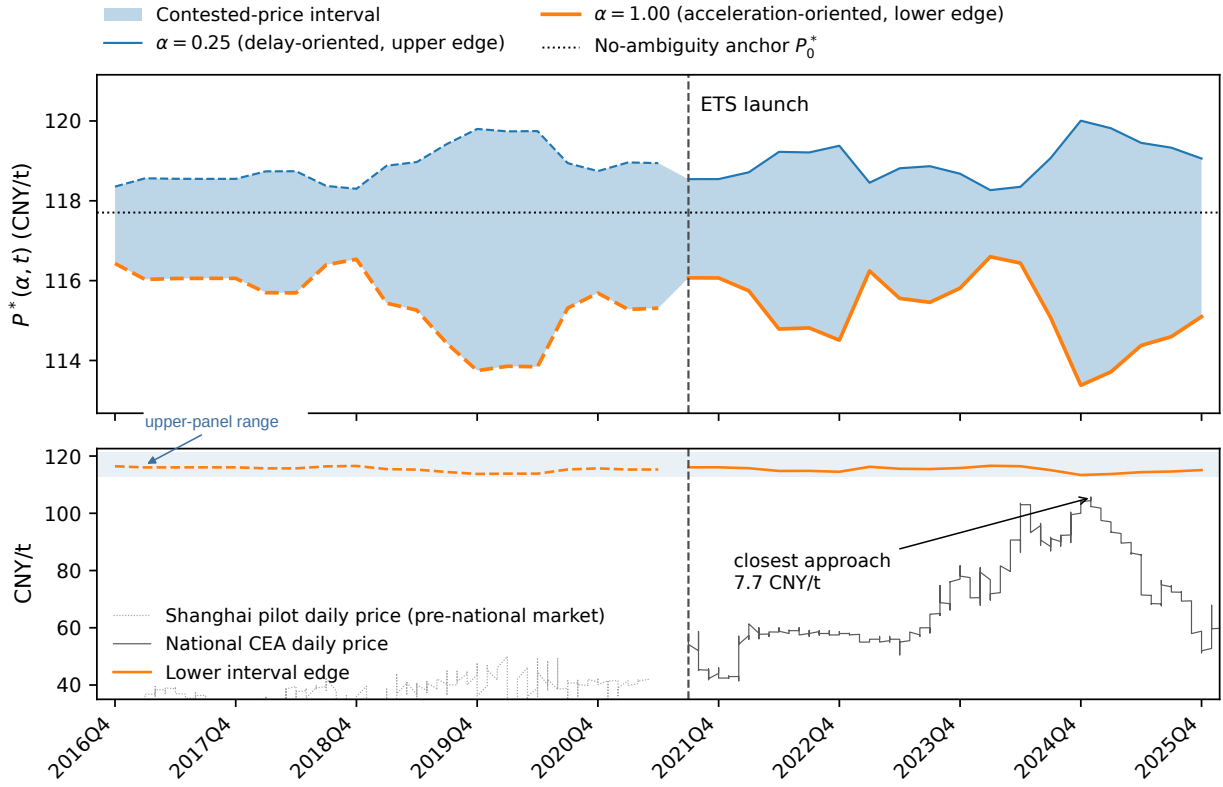


Figure 5. Retrospective fixed-anchor reconstruction of the contested-price interval for the four-member reference profile. Dashed threshold segments mark the period before the national ETS, and solid segments mark the national ETS period. The vertical dashed line marks the ETS launch, and the horizontal dotted line marks the anchor threshold. The lower panel plots archived daily prices from two markets: the Shanghai pilot price before the national market (dotted) and the national CEA price (solid). The closest-approach annotation compares the lower edge with the national CEA observations only. On the reported path, the lowest lower edge and the highest archived price both fell in 2024Q4, so their difference is the contemporaneous closest approach.

minimize. A non-positive lower-edge margin means at least one member classifies the price as invest. Unanimous investment also requires a non-positive upper-edge margin, with the price at or above the whole interval.

Under the baseline calibration, the highest archived price entered the interval at the 22% radius, the upper bound of the primary range. Among the 15 cells with a negative lower-edge margin, both margins were negative in four cells across two rows. Three of these lay in the $r = 0.06$ row, where the highest archived price exceeded the upper edge at radii up to 22%. At the 33% stress radius, the interval widened enough to contain this price. Two opposing channels shift the threshold level at this discount rate. The smaller characteristic root raises the project value trigger, whereas the thinner net discount margin lowers the factor converting triggers into prices. The discount-margin channel dominated, so every threshold fell. The fourth cell is Configuration A of Table 12, the lower volatility anchor at the frozen reference-peak radius. Beyond these perturbations, the upper-edge margin also depends on the estimation window of the volatility anchor. The thresholds were recomputed at the reference-peak radius under five national and two pilot windows, with either the relative or the absolute radius held fixed. The archived price stayed outside the interval in every window. The two pilot windows gave a volatility anchor of about 0.95 and

thresholds above 500 CNY/t.

The release rule of Eq. (18) raised an edge or member proximity flag in five of the 32 cells. Four of these lay at the 22% and 33% radii, where the mandatory wide-radius branch already covers all 16 cells. The fifth was Configuration A of Table 12. The 16 mandatory cells and this flagged cell gave 17 benchmark fallbacks. These fallbacks captured every disagreement between the two rules across the 32 cells. One gate-state disagreement occurred at the 33% radius under $r = 0.06$, where the affine rule reported a mixed gate and the benchmark reported unanimous invest. Two member-label disagreements also occurred. A separate diagnostic uses the 1.97 CNY/t price-recording buffer, which represents the recording granularity of the archived price series. Six of the 32 cells placed the highest archived price within this buffer of an interval edge. Three of these six cells were among the five flagged cells.

5.7. Further robustness and implementation requirements

Besides shifting the interval edges, the economic-parameter perturbations and the historical volatility anchors also moved the critical weight, as Table 14 reports. In every positive-radius case, the critical weight stayed above one half, and the member-threshold ordering was unchanged.

Table 12. Worked reports for two cells of the 32-cell sensitivity design, both at the $\bar{\sigma}_0 = 0.199$ calibration.

Report field	Configuration A ($\rho = 6.057\%$)		Configuration B ($\rho = 10\%$)	
	Affine	Benchmark	Affine	Benchmark
<i>Member-specific marginal carbon-price thresholds P_i^* (CNY/t)</i>				
$\alpha_i = 0.25$	105.65	105.60	106.74	106.62
$\alpha_i = 0.50$	104.13	104.07	104.23	104.07
$\alpha_i = 0.75$	102.61	102.56	101.73	101.60
$\alpha_i = 1.00$	101.09	101.09	99.22	99.22
<i>Interval, prevailing price, and classification</i>				
Critical weight $\tilde{\alpha}$ (affine)	0.510		0.517	
Lower edge of C_t (CNY/t)	101.09	101.09	99.22	99.22
Upper edge of C_t (CNY/t)	105.65	105.60	106.74	106.62
Width B_t (CNY/t)	4.56	4.51	7.53	7.41
Prevailing price P_t (CNY/t)	105.65		105.65	
Members classified invest	4 of 4	4 of 4	3 of 4	3 of 4
Gate state	unanimous invest	unanimous invest	mixed	mixed
<i>Release rule of Eq. (18), evaluated on the affine report</i>				
d_{edge} (CNY/t)	0.002		1.092	
d_{member} (CNY/t)	0.002		1.092	
Registered envelope E_R (CNY/t)	0.059		0.160	
Price-recording buffer flag, distance within the 1.97 CNY/t buffer (diagnostic only)	unresolved by price record		unresolved by price record	
Proximity flags	edge, member		none	
Reporting mode	dual report		affine-only	
Released classification	unanimous invest		mixed	

Notes. The calibration is $r = 0.08$, $\mu_c = 0.04$, and $\bar{\sigma}_0 = 0.199$, with the reference project value $V_0 = I$ and the four-member reference profile. P_t is the highest archived national CEA closing price over 2021Q3 to 2025Q4. Configuration A uses the frozen archived canonical serialized radius key, and its displayed value is rounded for this table only. The upper edge of Configuration A was 105.6481 CNY/t before rounding, and the prevailing price was 105.65 CNY/t. The edge therefore lay below the price, although both display as the same rounded value. Edges and widths are rounded independently, so differences of displayed edges can deviate from the displayed width in the last digit. Affine and benchmark thresholds coincide at $\alpha_i = 1.00$ by construction. In dual mode, a classification is released only where the two rules agree. In affine-only mode, the benchmark column is shown for comparison.

Table 13. Signed margin of each edge of the contested-price interval relative to the highest archived carbon price. The configurations are the economic perturbations and volatility anchors of the sensitivity analysis. SHEA denotes the Shanghai Emission Allowance pilot market.

Configuration at $h/\bar{\sigma}_0$	Margin to the lower edge (CNY/t)				Margin to the upper edge (CNY/t)			
	6.057%	10%	22%	33%	6.057%	10%	22%	33%
Baseline calibration	+7.7	+5.0	-2.8	-9.3	+14.4	+16.0	+21.3	+26.8
$r = 0.06$ (-25%)	-16.8	-19.3	-26.2	-31.9	-10.8	-9.4	-4.4	+0.9
$r = 0.12$ (+50%)	+55.0	+51.9	+42.7	+34.9	+62.7	+64.5	+70.4	+76.4
$\mu_c = 0.06$ (+50%)	+0.5	-1.9	-8.7	-14.2	+6.5	+8.0	+13.0	+18.3
$\bar{\sigma}_0 = 0.199$ (-20%)	-4.6	-6.4	-11.8	-16.3	-0.0019	+1.1	+4.7	+8.5
$\bar{\sigma}_0 = 0.299$ (+20%; early CEA anchor regime)	+21.8	+18.1	+7.5	-1.2	+30.9	+33.1	+40.4	+48.1
Pooled national, 2021Q3 to 2025Q4 ($\bar{\sigma}_0 = 0.278$)	+15.7	+12.4	+3.0	-4.7	+23.7	+25.6	+32.1	+38.8
SHEA pilot, 2016 to 2021Q1 ($\bar{\sigma}_0 = 0.953$)	+376.9	+345.6	+258.3	+188.5	+455.1	+474.7	+542.3	+614.9

Notes. Each margin is the stated interval edge minus the highest archived national CEA closing price over 2021Q3 to 2025Q4 (105.65 CNY/t, 1,073 observations). A positive lower-edge margin means every member classifies every archived price as wait. A negative lower-edge margin with a positive upper-edge margin means a mixed gate. When both margins are negative, every member classifies the price as invest. The radius is fixed within each column. The 6.057% value is the displayed peak of the reconstructed signal path. This radius uses the frozen archived canonical serialized key, and its displayed value is rounded for the table only. The 10% radius is the affine-only cap, 22% is the upper bound of the primary radius range, and 33% is the stress radius. Classifications and release decisions use unrounded values. A margin below 0.05 CNY/t in absolute value is shown at four decimals, so its sign remains legible. At a given radius, the upper-edge margin minus the lower-edge margin equals the interval width up to rounding. It equaled 6.63 CNY/t only in the baseline row at the reference-peak radius. Each configuration matches the corresponding row of Table 14. The underlying edge values are in data/observed_margin_both_edges.csv of the replication archive.

Implementation safeguards. An elicitation instrument for the recorded weights must meet three requirements. It must return a number in $[0, 1]$ on the adverse endpoint of the declared ambiguity set. A calibration to a different object needs a stated mapping. The instrument must be reproducible at the same gate, because the recorded weight enters the gate audit record. Its precision must also be assessed against the review band around the critical weight. A prospective implementation should therefore prespec-

ify standardized administration with attention and comprehension checks. It should also prespecify a test-retest assessment of rank-order reliability and a check of scale comparability across gates. Candidate instruments include matching-probability tasks in the style of Dimmock et al. (2016), lottery equivalents, and structured vignettes. A discrete-choice format such as best-worst scaling could also serve, with its output mapped onto $[0, 1]$ by a rule fixed in advance. The instrument version, acceptance

Table 14. Sensitivity of the critical weight under the volatility-only ambiguity declaration.

Panel A. Economic-parameter sensitivity at a 10% relative radius				
Perturbation		$\hat{\alpha}$ at 10%		
Baseline calibration		0.517		
$r = 0.06$ (−25%)		0.520		
$r = 0.12$ (+50%)		0.514		
$\mu_c = 0.06$ (+50%)		0.521		
$\bar{\sigma}_0 = 0.199$ (−20%)		0.517		
$\bar{\sigma}_0 = 0.299$ (+20%)		0.518		
Panel B. Historical-anchor sensitivity across three relative radii				
Historical anchor regime	Anchor estimate	10%	22%	33%
Shanghai Emission Allowance (SHEA) pilot, 2016 to 2021Q1	$\bar{\sigma}_0 = 0.953$	0.524	0.553	0.580
Early CEA, 2021Q3 to 2023Q4	$\bar{\sigma}_0 = 0.299$	0.518	0.540	0.560
Current CEA, 2024 to 2025 (baseline)	$\bar{\sigma}_0 = 0.249$	0.517	0.538	0.557
Pooled national, 2021Q3 to 2025Q4	$\bar{\sigma}_0 = 0.278$	0.518	0.539	0.558

Notes. The $\bar{\sigma}_0 = 0.299$ perturbation of Panel A coincides numerically with the early CEA anchor regime of Panel B.

rule, and recorded weights should be frozen before later gate outcomes are observed. An extreme weight sets an interval edge, so it should prompt a documented review before any exclusion. Two further choices belong to institutional policy, and the first is reporting granularity. The edges, the width, and the three-state summary are symmetric functions of the member thresholds, so a sorted list without member identities preserves them. A sorted list, however, removes the link between each threshold and its member. An identified table keeps this link, although it also exposes the most divergent member in the minutes, as the documented review of extreme weights does. The second choice is the elicitation schedule, which matters because the extreme recorded weights alone determine both edges. A member aware of this mapping could move the reported interval through the choice of weight, so the method assumes weights recorded in advance. Collecting the weights before project materials circulate establishes a pre-disclosure reference, provided the elicitation names the common information behind them. A project-specific adjustment of the weights would require a second elicitation under a documented information set.

The acceptance rule for the instrument can rest on a precision target derived from the model. Under the classical error model, measurement noise with standard deviation σ_α enters each recorded weight. A member whose recorded weight lies a distance d from the critical weight then receives the opposite directional label with probability $\Phi(-d/\sigma_\alpha)$. Holding this probability at or below p requires $\sigma_\alpha \leq d/\Phi^{-1}(1-p)$. With d set to the illustrative review band of 0.05, the calculation gives $\sigma_\alpha \leq 0.039$ for a 10% flip probability and $\sigma_\alpha \leq 0.030$ for 5%. These values serve as design targets for the ordinal use of the weights. Members closer than d to the critical weight keep de-emphasized labels, since no instrument precision secures a stable label arbitrarily close to it. Beyond noise within a gate, recorded weights can also drift between gates. When the recorded weights of simulated panels drifted by 0.05 per gate, the three-state summary changed in 28.9% of the panels. The re-elicitation cadence is therefore a design parameter.

These elicitation choices join the other configuration deci-

sions a prospective gate must record, which Table 15 organizes by the three layers of Figure 1. The text-pipeline items and the constant c_w serve only a radius derived from documents, so a committee declaring its radii directly drops them. Its threshold report stays reproducible from the recorded radius, although the expert reasoning behind this radius cannot be regenerated from a text pipeline. In the interface layer, the registry snapshot authorizes affine-only release, so an institution always issuing the dual report needs no registry. It then loses both the option to omit the benchmark and the record of the configuration authorizing a past release. It can still retain and replay the inputs and outputs of each gate.

The irreducible configuration is shorter than Table 15 suggests. At minimum, a prospective configuration fixes four items: the economic primitives, the volatility anchor, the declared coordinate, and the weight-recording protocol. Each gate then supplies its declared radius, the recorded member weights, and the prevailing price. Benchmark recomputation also requires the registered reference project value. The declared coordinate cannot be dropped, because the directional labels are unreadable without it. Where no member weights are recorded, a hypothetical profile still yields a well-defined interval, which serves as a sensitivity band over a stipulated grid.

5.8. Preconditions on the reported basis

Beyond the recorded configuration, five preconditions govern whether the gate can accept the basis of the reported interval. Each precondition names an accounting choice, a model assumption, or an input-validity question, and each is documented before the report is interpreted.

The first precondition is the cost basis, because the reported threshold is marginal. A committee comparing it with a figure including operating costs first applies the common adjustment. Across the 18 national ETS gates with archived prices, no net cost changed the reported classification, because these prices already lay below the interval. A net subsidy changed the classification only at tens of CNY/t, a scale specific to the reference configuration. One gate turned mixed at $\Delta = -10$ CNY/t. At $\Delta = -50$ CNY/t, ten of the 18 gates read unanimous invest, and eight remained unanimous wait. At $\Delta = -80$ CNY/t, all 18 read unanimous invest. The reported thresholds also carry basis risk against the realized carbon cost of a plant. Two common cost components map onto the adjustments of the reported basis. A per-tonne transport and storage charge is the additive adjustment Δ . An energy penalty scales the captured volume by a common factor at fixed utilization, so it is a multiplicative adjustment.

The second precondition is the horizon. Replacing the perpetuity with a finite-horizon annuity multiplies every member threshold, the interval width, and every price-unit envelope by one common factor. The member ordering, the directional labels, and the relative spread survive, while the classification of a given price may change. A gate reporting on an annuity basis therefore rescales the envelope with the thresholds. A finite horizon can also enter the stopping problem. Between a five-year horizon and the perpetuity, it moved the critical weight by less than 0.03 at every examined radius.

Table 15. Configuration record required before a prospective gate, organized by the three layers of Figure 1.

Configuration item	Registered before the gate	Opens a new dated cell	Left to the deploying institution
<i>Input layer</i>			
Volatility anchor $\bar{\sigma}_0$	Estimation window, market, and estimator	New window, market boundary, or estimator	An option-implied or otherwise forward-looking anchor
Signal-to-radius constant c_w	Target radius h_{ref} , reference signal w_{ref} , and $c_w = h_{\text{ref}}/w_{\text{ref}}$	Any change to either input, or to the encoder	An empirical calibration, unidentified by the available price record
Text pipeline	Encoder, layer, pooling, normalization, truncation, corpus screen	Any change to any of these items	Evidence of semantic dispersion predicting realized volatility
Recorded weights $\{\alpha_i\}$	Instrument version, mapping interval, review band, and acceptance rule. The rule must meet the precision target $\sigma_\alpha \leq 0.039$ at $d = 0.05$ and $p = 0.10$	New instrument or mapping interval	Universal acceptance cutoffs and a committee pilot
<i>Analytical layer</i>			
Economic primitives	r, μ_c, I, Q , and the admissibility margin ϵ	Any change	Audited full-cost components for a viability appraisal
Reference value V_0/I	The value at which the benchmark is evaluated	Any change	Release authorization at $V_0/I \neq 1$
Declared coordinate	Standard deviation or variance, named explicitly	Any change	None
<i>Interface layer</i>			
Registry snapshot	Version v , canonical serialized keys, envelope version, solver version	Any change to any identifier	A continuous-radius or global error certificate
Release and reporting	Review cadence, dual-report format, directional-label conventions	Schedule or format change	A voting rule, a consensus procedure, or a full project appraisal

Notes. Every item is frozen before later gate outcomes are observed. Release matching uses the frozen serialized key from the registered pipeline. The fourth column lists the items a deploying institution supplies.

The third precondition is the ambiguity channel, since the model places ambiguity on volatility alone and holds drift fixed. At a 10% volatility radius, adding drift ambiguity of the same relative size multiplied the interval width by 2.41. It also raised the maximum relative trigger error from 0.26% to 1.48%. This joint value at a 10% radius already exceeded the 1.26% maximum of the volatility-only declaration at radii up to 22%. A gate under a joint declaration therefore re-evaluates accuracy before release.

The fourth precondition is the price process. On the 1,073 archived log prices, an augmented Dickey-Fuller regression did not reject a unit root, with or without a trend. The Lo-MacKinlay variance-ratio test did not reject a random walk at two, five, ten, or twenty days. These tests did not reject the persistence assumed by the geometric Brownian motion for the price level. Gaussian increments, however, were rejected for the returns of the same series. The implied mean-reversion speed was 0.818 per year, but its confidence interval contained zero so the series did not identify a speed. The point estimate still lay above the largest reversion speed in Table 3. On this grid, a positive speed moved the anchor threshold from 116.88 CNY/t to between 97.16 and 99.81 CNY/t. The anchor threshold of 116.88 CNY/t was recomputed on the same Crank-Nicolson grid so the comparison isolates mean reversion from the solver's level bias. The level shift alone exceeded the 7.73 CNY/t reference margin, so a gate expecting mean reversion should recalibrate. Moving half the declared variance into jumps, at the same total, had a smaller effect. The anchor threshold moved from 116.88 to 115.41 CNY/t, the interval width changed by at most 1.04% and the critical weight rose.

The fifth precondition is the radius input, because the mapping of the policy-text series to realized volatility was unstable

in the available record. A registered constant therefore sets the radius scale, and the reference result is sensitive to this constant. An alternative radius based on the sampling uncertainty of the volatility anchor was 6.37%, close to the 6.057% peak of the reference path. It left the archived price outside the interval. A radius based on the dispersion of quarterly realized volatility was about seven times larger and made the gate mixed. Every evaluated input alternative changing the classification lay outside the 10% cap.

Outside the CCUS setting, a transfer to a new application takes six steps, and the first three rebuild the theory. The first defines a common ordered uncertainty set and the meaning of the recorded weight on its adverse endpoint. The second derives the stopping object and verifies the existence of a unique scalar boundary. The third identifies the adverse endpoint as the endpoint with the lower trigger and then rederives trigger monotonicity, curvature, and the critical weight. Three further steps rebuild the reporting apparatus. The fourth maps the member triggers to a common operational scale. The fifth reconstructs the finite benchmark comparison and the versioned release registry. The sixth validates the input, the elicitation, and the scale before deployment. The workflow carries over across applications, whereas the formulas and signs obtained here need rederivation or checking.

The CCUS reconstruction points to three patterns. First, the reference configuration kept every gate at unanimous wait, whereas changing one threshold-side input could produce binding configurations. Second, the common appraisal inputs moved the interval edges more than member disagreement did. Third, the five preconditions on the reported basis shaped the reading of the thresholds. These patterns come from one retrospective reconstruction and may differ in other projects. They nonetheless

less indicate where a committee might focus before reading the member-threshold vector.

6. Conclusions

This study addressed one stage-gate reporting problem. Members sharing the same project information may weight adverse scenarios differently, and a compressed report hides whether their classifications of the prevailing price diverge. The analytical results, the benchmark evaluation, and the CCUS reconstruction jointly support one central finding. Within the maintained model, the member-threshold vector and the contested-price interval showed at each gate whether all members classified the prevailing price alike. A unique critical weight partitions these thresholds around the no-ambiguity anchor, and trigger curvature determines its location.

Trigger curvature changes sign with the declared ambiguity coordinate, so the affine critical weight falls on a coordinate-dependent side of one half. Under the pre-commitment benchmark, this side also depends on the reference project value. In standard-deviation units, the affine critical weight lies above one half throughout the domain. A symmetric declaration in variance units at the same relative radius selects different endpoint volatilities and reverses this placement. Evaluating the pre-commitment benchmark at $V_0 = I$ also reversed it.

Beyond the direction of each member shift, the benchmark comparison measured the size of the affine error. Across the 33 admissible calibrations, the maximum member-trigger error was 2.74% of the benchmark over the primary radius range up to 22%. This maximum reached 6.15% with the 33% stress radius included. Small trigger errors could still change classifications, and the two classification-fidelity criteria held only at the evaluated radii up to 10%. This radius therefore caps affine-only release. On held-out configurations, the price-axis release distance behaved as on the evaluated grid, whereas both normalized criteria failed on one profile clustered near one half. There, a similar absolute displacement formed a much larger fraction of the narrow benchmark interval. The release rule therefore conditions on absolute distances.

In the CCUS application, the unanimous-wait outcome depended on the configuration. Under the reference configuration, none of the 1,073 archived allowance prices reached the interval in force at the time. One classification summary would therefore have sufficed at every gate. With the price record fixed and only the threshold side perturbed, this outcome did not hold in 15 of the 32 sensitivity configurations. Eleven of these 15 cells went to benchmark recomputation, ten of them at radii above the affine-only cap. The remaining four cleared the release rule on the affine report alone. These results have two practical implications. First, member disagreement moved the reported edges less than the common appraisal inputs did in this illustration. A committee with an unsettled cost basis would thus weigh member attitudes inside an interval narrower than the shift from a 10% cost revision. Second, repeated unanimous classifications may arise even with dispersed member thresholds, because the prevailing price can stay outside the contested-price interval. Unanimity alone therefore says little about the closeness of the

thresholds. These implications hold under the maintained model, the reference profile, and the declared ambiguity coordinate.

The most important limitation is the maintained price model. Every member threshold is derived under a perpetual geometric Brownian motion. This assumption matters because the threshold levels and the interval edges depend on the price process. Under mean reversion, the anchor threshold moved by more than the 7.73 CNY/t margin between the highest archived price and the lower edge. The limitation is acceptable here because unit-root and variance-ratio tests on the archived prices did not reject the persistence assumed by the model. In the robustness grids, the critical weight also stayed above one half under mean reversion, a finite horizon, and jumps at fixed declared variance. A gate expecting another process recalibrates before release.

Within these limits, the proposed report moves the detection of member disagreement from the vote to the screening step before it. By using one recorded weight per member, it allows the committee to determine before any aggregation whether one classification summary suffices. Where the members disagree, the report locates the disagreement in price units, and at every gate it states whether the affine rule can stand alone.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The anonymized replication archive accompanying this submission contains the code and the processed data, including the price series and the derived document vectors. Its README itemizes the contents, and a per-file SHA-256 manifest lists every file. The 74 commands listed in the README were run from a clean copy, and 68 of them completed. Of the rest, 5 were skipped because their inputs are not redistributed, and one legacy diagnostic returned its expected nonzero status. The code is released under the MIT License and the processed data under CC BY 4.0. The archive is a static snapshot of the evaluated registry cells and the fallback logic. The 682 policy documents are third-party material from the PKULaw database and are not redistributed. The archive lists the stable PKULaw document number of each record, so a reader with database access can retrieve the same set. One environment variable then points the pipeline to the local copy of these documents. The carbon prices are published CEA and Shanghai pilot closing prices. The encoder weights come from publicly released models used under their own licenses. The submission-stage archive has no permanent external repository identifier yet.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used a generative AI assistant for language editing, sentence restructuring,

and manuscript organization. After using this tool, the authors reviewed and edited the content as needed. They independently verified the scientific claims, the derivations, the numerical results, and the references. The authors take full responsibility for the content of the article. The authors specified and executed the models, data processing, code, and analyses. The replication archive documents every command producing a reported number.

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